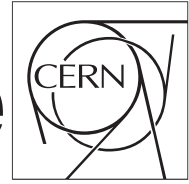




The Compact Muon Solenoid Experiment

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Observation of elliptic flow of prompt J/ψ in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

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Abstract

Measurement of elliptic flow of prompt J/ψ meson in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV is reported with the CMS experiment at the LHC. The J/ψ mesons are reconstructed via the dimuon channel at the forward rapidity region ($-2.86 < y_{cm} < -1.86$ and $0.94 < y_{cm} < 1.94$). Significant positive v_2 values, extracted from long-range two-particle correlations, are observed for prompt J/ψ over the transverse momentum range of $2 < p_T < 8$ GeV in high-multiplicity (about 200 charged particles) pPb events. This observation provides a direct evidence for the presence of charm quark collectivity developed in small collision systems. By comparing with previous CMS measurements of elliptic flow for open charm mesons (D^0) and strange hadrons, strong constraints can be obtained on the collective dynamics of charm quarks produced in small hadronic systems with high multiplicities.

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1 Introduction

The observation of strong collective behavior in the azimuthal correlations of particles emitted over a wide pseudorapidity range in relativistic nucleus-nucleus (AA) collisions at the BNL RHIC [1–4] and later at the CERN LHC [5–10], has suggested the formation of a strongly interacting quark-gluon plasma (QGP) which exhibits nearly ideal hydrodynamic behavior [11–13]. In recent years, similar long-range collective azimuthal correlations are also observed in events with high final-state particle multiplicity resulting from proton-proton (pp) [14–17], proton-lead (pPb) [18–26], and lighter nucleus-nucleus systems [27–29], which raised the question whether a fluid-like QGP medium is created in these much smaller collision systems. In the context of hydrodynamic models, the azimuthal correlation structure of emitted particles is typically characterized by its Fourier components [30]. In particular, the second and third Fourier anisotropy components are known as elliptic (v_2) and triangular (v_3) flow, which most directly reflect the QGP medium response to the initial collision geometry and its fluctuations [31–33]. While experimental measurements in these small systems are consistent with a hydrodynamic expansion of a tiny QGP droplet, alternative scenarios based on gluon saturation in the initial state also claim to capture the main features of the correlation data (see Ref. [34] for a review).

Heavy flavor quarks (charm and bottom) are produced in the early stages of the hadronic collision via hard scatterings, due to their large masses. In AA collisions, as they are decoupled from the bulk production of soft gluons and light flavor quarks, they have the opportunity of probing the properties and dynamics of the QGP through its entire history of evolution [35]. Strong elliptic flow signal of open heavy flavor D^0 mesons has been observed in both AuAu collisions at RHIC [36] and PbPb collisions at the LHC [37–39], suggesting that charm quarks develop strong collective flow behavior, similar to the bulk of the QGP medium. Furthermore, measurement of elliptic flow of hidden charm mesons, J/ψ , provided stronger evidence for the fast thermalization of charm quarks and their subsequent collectivity via strong rescatterings in the deconfined QGP medium [40].

A key open question in the study of collectivity in small systems, such as pp and pPb collisions, is whether strong collective behavior also extends to heavy flavor quarks. Recent observation of significant elliptic flow signal for prompt D^0 meson in pPb collisions provided the first strong evidence for charm quark collectivity in small systems. The observed v_2 signal for D^0 meson is found to be smaller than that of light flavor hadrons at a given particle transverse momentum (p_T), indicating a weaker collective motion for charm quarks developed there than that in large AA systems. However, as the D^0 meson carries both a light and a charm quark, their relative contributions to the observed v_2 signal is not fully constrained. Without detailed theoretical modeling, a scenario of the D^0 meson v_2 entirely driven by the light flavor quark v_2 is not excluded (although unlikely based on the observed magnitude of D^0 v_2). Observation of an elliptic flow signal for J/ψ meson in small systems will provide the most direct evidence of charm quark collectivity, and impose new constraints on the collective dynamics of heavy quark production in small systems. A first attempt of measuring inclusive J/ψ v_2 in pPb collisions is reported in Ref. [41], where an indication of positive signals is seen for $3 < p_T < 6$ GeV. If indeed a QGP medium is formed in small systems, observation of heavy quark collectivity may also provide evidence of interactions between hard probes and the medium, a key observation in large AA systems that is currently lacking in small systems.

This Letter presents the measurement of the elliptic flow of prompt J/ψ mesons (excluding contribution from B meson decay) from long-range two-particle correlations in very-high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV. The v_2 harmonic for prompt J/ψ mesons at

77 forward rapidity ($-2.86 < y_{\text{cm}} < -1.86$ and $0.94 < y_{\text{cm}} < 1.94$) is determined over a wide
 78 p_{T} range from 0.2 GeV to 10 GeV. To estimate the possible residual contribution from back-
 79 to-back jet-like correlations, the v_2 results are also presented after subtracting correlations ob-
 80 tained from low-multiplicity pPb events (denoted as v_2^{sub}). The results are compared to those
 81 of strange and prompt D^0 mesons, measured in the same kinematic ranges, as a function of
 82 p_{T} and transverse kinetic energy per constituent quark ($KE_{\text{T}}/n_{\text{q}}$, where $KE_{\text{T}} = \sqrt{m^2 + p_{\text{T}}^2} - m$,
 83 and n_{q} is the number of constituent quarks) to explore collectivity at parton level.

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2 Data and Monte Carlo samples (Done)

2.1 pPb collision

The analysis of J/ψ v_2 in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV is performed using data recorded by the CMS experiment at the CERN LHC, correspond to a total integrated luminosity of about 186 nb^{-1} . Official golden json file has been applied to select runs and lumi sections certified by the CMS data certification team. CMSSW release 8.0.24 was used for prompt reconstruction. Primary data sets used can be found in Table 1.

Table 1: Official data samples for pPb at $\sqrt{s_{NN}} = 8.16$ TeV

Run Range	Dataset
285479–286496	/PAHighMultiplicity1-6/PARun2016C-PromptReco-v1/AOD
285479–286496	/PADoubleMuon/PARun2016C-PromptReco-v1/AOD

Monte Carlo (MC) samples of pPb collisions are generated in this analysis to determine the performance of track reconstruction, event selection efficiency and cross-check of other potential detector effects. Samples with J/ψ particle generated are also used to study reconstruction performance. The samples are produced with the “data-taking” detector conditions, and are listed in Tables below.

Table 2: MC samples for 8.16 TeV pPb collisions with EPOS and HIJING generators

Generator	Dataset
HIJING, MB	/HIJING_pPb_8160_DataBS/pPb816Summer16DR-MB_80X_mcRun2_pA_v4-v2/AODSIM
HIJING, MB	/HIJING_PbP_8160_DataBS/pPb816Summer16DR-MB_80X_mcRun2_pA_v4-v2/AODSIM
EPOS, MB	/ReggeGribovPartonMC_EposLHC_pPb_4080_4080_DataBS/pPb816Summer16DR-MB_80X_mcRun2_pA_v4-v2/AODSIM
EPOS, MB	/ReggeGribovPartonMC_EposLHC_PbP_4080_4080_DataBS/pPb816Summer16DR-MB_80X_mcRun2_pA_v4-v2/AODSIM
PYTHIA, J/Ψ	/Psi1SToMuMu_pTMu-2p5_PbP-Bst_8p16-Pythia8/pPb816Summer16DR-PbPBst_80X_mcRun2_pA_v4-v1/AODSIM
PYTHIA, J/Ψ	/Psi1SToMuMu_pTMu-2p5_PbP-Bst_8p16-Pythia8/pPb816Summer16DR-pPbBst_80X_mcRun2_pA_v4-v1/AODSIM
PYTHIA, NP J/Ψ	/NonPrPsi1_2SToMuMu_pTMu-2p5_PbP-Bst_8p16-Pythia8/pPb816Summer16DR-PbPBst_80X_mcRun2_pA_v4-v1/AODSIM
PYTHIA, NP J/Ψ	/NonPrPsi1_2SToMuMu_pTMu-2p5_PbP-Bst_8p16-Pythia8/pPb816Summer16DR-pPbBst_80X_mcRun2_pA_v4-v1/AODSIM

3 Triggering and Event Selections

3.1 Di-Muon trigger

Events collected by the HLT_PAL1DoubleMuOpen.v1 trigger are used as the primary data sample for estimating potential contamination of back-to-back jet-like correlations. This trigger requires two muon candidates found in the muon detectors at level-1 (L1) trigger with loosest possible selections to maximize the detection efficiency. During 2016 pPb run, this trigger was operated without any prescale, and thus provides high statistical precision for correlation studies than Minimum-Bias (MB) triggered events (prescaled by about 100 on average) in the low multiplicity region.

3.2 High multiplicity triggers

In the high-multiplicity region, dedicated high-multiplicity triggered pPb events are used, same as previous analyses such as HIN-16-022 and HIN-17-003. We describe the details here again to remind the readers.

For proton-lead collisions at 8.16 TeV, the events in the charged multiplicity ranges above 120 were selected online using dedicated triggers with charged particle multiplicities thresholds of: 120, 150, 185, 250, and 280. The first two thresholds were seeded by Level-1 Minimum Bias triggers, while the others were seeded by tower count (TC) trigger in barrel ECAL and HCAL detectors, where events were selected if passing a criterium in the number of active towers. An active tower was defined as a trigger tower (ECAL+HCAL) with a transverse energy greater than 0.5 GeV. It was used $TC > 115$ or 120 for multiplicity between 185 and 250 and $TC > 145$ or 150 for multiplicity above 250. The reason for the usage of two TC thresholds for the same multiplicity range is related with the observation of a considerable variation in the noise level of HCAL during data-taking, which showed some dependence with the LHC filling scheme for the energy considered for TC. This non-uniformity in the trigger performance was taken into account in this analysis with the estimation of corresponding systematic uncertainties (See Sec. 8).

The multiplicity thresholds at HLT step are defined on a set of tracks selected with:

- $p_T > 0.4$ GeV
- $|\eta| < 2.4$
- $\text{MinSepZ}_{\text{pixel}} < 0.12$
- $\text{MinSepZ}_{\text{full}} < 0.15$.

Where $\text{MinSepZ}_{\text{pixel}}$ and $\text{MinSepZ}_{\text{full}}$ represent the minimum separation in z between the track and vertex for tracks reconstructed using only the pixel detector and using full tracker, respectively. In addition, an event selection based on the number of pixel tracks (ranging from 40 to 140, depending on the full-tracker track multiplicity thresholds) and vertex distance in z with respect to the detector origin smaller than 15 cm is applied.

To measure the triggers efficiency a reference method was used, where a minimum bias trigger is used as reference. The efficiency is computed using the following expression:

$$\text{Trigger Efficiency} = \frac{\text{N Events Firing Desired Trigger \& MinBias Trigger}}{\text{N Events Firing MinBias Trigger}}.$$

The Fig. 1 shows the efficiency turn-on curves for the L1 trigger component (Tower Count),

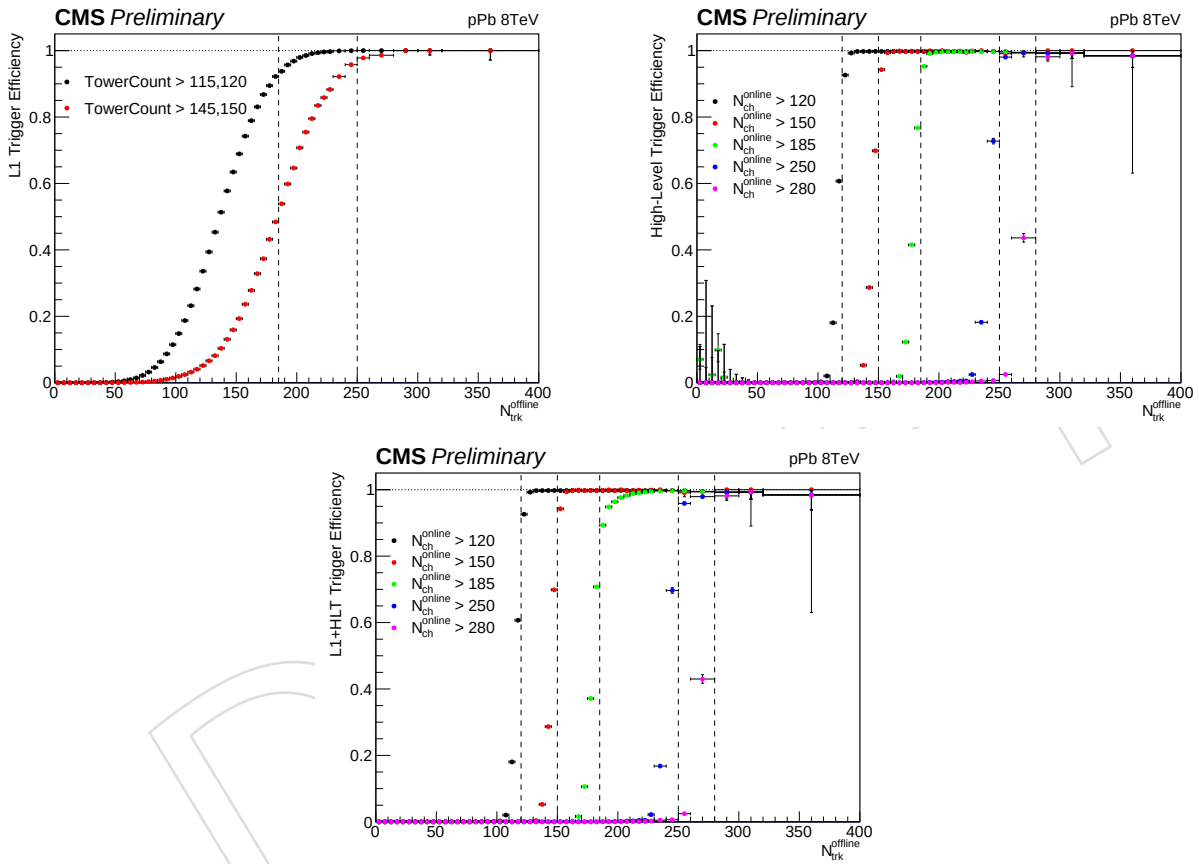


Figure 1: High multiplicity trigger efficiency as a function of offline selected track multiplicity. Top left: level-1 trigger. Top right: HLT-only component. Bottom: total efficiency, L1+HLT. The vertical lines indicate the online track multiplicity selection for each trigger.

HLT-only component, and the total trigger (L1 + HLT) as a function of offline track multiplicity ($N_{\text{trk}}^{\text{offline}}$). In general, an efficiency plateau above 95% is observed in the $N_{\text{trk}}^{\text{offline}}$ range for the analysis what corresponds to our performance target for the HM triggers.

Total number of events recorded for each HLT trigger path can be found in Table. 3.

Table 3: Total number of pPb events recorded during HI run 2016 at TeV $\sqrt{s_{NN}} = 8.16$ TeV for each HLT path

HLT path	Number of events
HLT_PAFullTracks_Multiplicity120	286.51M
HLT_PAFullTracks_Multiplicity150	216.14M
HLT_PAFullTracks_Multiplicity185*	795.7M
HLT_PAFullTracks_Multiplicity250	64M
HLT_PAFullTracks_Multiplicity280	13.36M

In this analysis, events triggered by HLT_PAFullTracks_Multiplicity185 with event multiplicity $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ are used. Here, event multiplicity, $N_{\text{trk}}^{\text{offline}}$, is defined as number of tracks with $|\eta| < 2.4$ and $p_T > 0.4$ GeV/c. Around 550 million events are used in this analysis.

3.3 Offline event selections

The offline event selections are identical to all other pPb 8 TeV analysis, e.g. AN-16-427 (HIN-16-021), AN-16-370 (HIN-16-010). Selected details are shown below.

3.3.1 Offline selection

There are several background sources to the inelastic hadronic collisions such as beam-gas collisions and beam scraping events. In addition the event samples contain electromagnetic interactions (ultra-peripheral collisions). In order to extract a pure hadronic collision sample for analysis, offline event selection criteria were applied to the minimum bias triggered event sample:

- Beam-scraping filter: the PKAM background events are characterized by large multiplicities observed in the tracker without any sign of a common origin of the tracks within the CMS Detector. In order to remove these large multiplicity events a so-called beam scraping filter was developed for pp analyses. The filter removes events where the number of reconstructed tracks is larger than a certain threshold but the fraction of good quality tracks is low. The requirement of this filter are:
 1. The minimum number of tracks required to check the quality of the event is 10.
 2. The minimum fraction of good quality tracks in the event is required to be larger than 25%.
 3. The good quality on the tracks is the highPurity requirement.
- offline, the presence of at least one tower with energy above 3 GeV in one or both of the HF calorimeters (HFO conditions). A valid reconstructed vertex is required to be within $|v_z| < 15\text{cm}$ and the number of tracks associated with this vertex has to be greater than 2.
- HF coincidence filter: this filter requires at least 1 tower on each side of the interaction point in the HF calorimeter with an energy deposit above 3 GeV threshold. This requirement removes approximately 99 % of UPC events.

3.3.2 Background studies with unpaired bunches

To study the possible contamination from beam-scraping, beam halo and beam backgrounds, zerobias samples with paired and unpaired bunches are also recorded and analyzed.

A study of non-colliding bunches, shown in Fig. 2, shows that the primary vertex requirement is sufficient to reject all backgrounds not originating from proton-lead collisions. The probability to select events in the presence of a single (noncolliding) beam is about 2×10^{-5} per bunch crossing. Consequently, the contribution of background events from beam, beam halo, and cosmic ray sources to the observed yields is negligible.

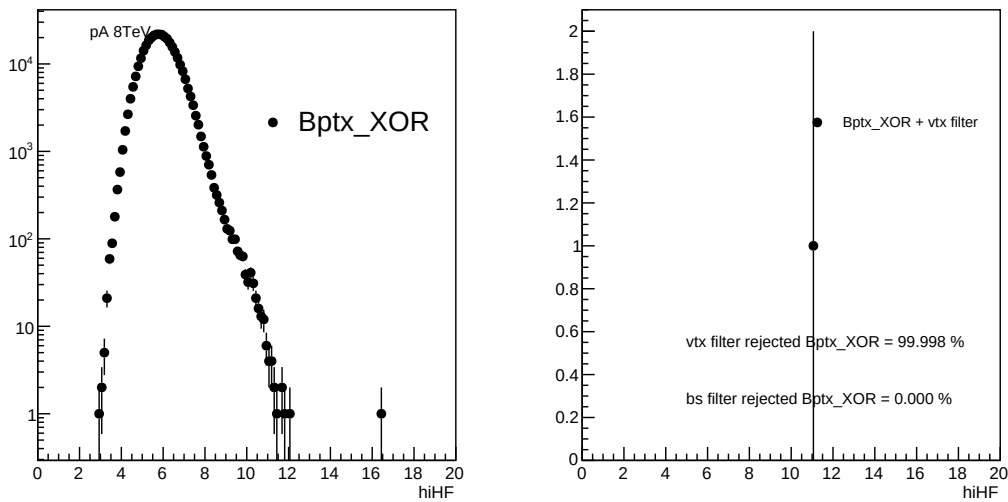


Figure 2: Left: Unpaired bunches (BPTX XOR) as a function of the sum of transverse energy in the forward hadronic calorimeter (hiHF), measured in GeV. Right: Same distribution after vertex selection requirement. The values of the fraction of background events removed by the beam-scraping and vertex filters are written in the plot.

3.3.3 Contamination from electromagnetic events after event selection

The remaining EM contamination after the coincidence filter requirement can be estimated from data, by using the filter efficiency computed in MC. For that we use a sample of events with a tighter selection, requiring at least two towers per side with an energy above 3 GeV. The standard sample (with one tower per side requirement) and the sample with tighter selection are weighed in an event-by-event basis by the efficiency of the corresponding filters obtained in MC. The EM contamination can be estimated from the difference of the number of events in the two samples, and is about 1% or less.

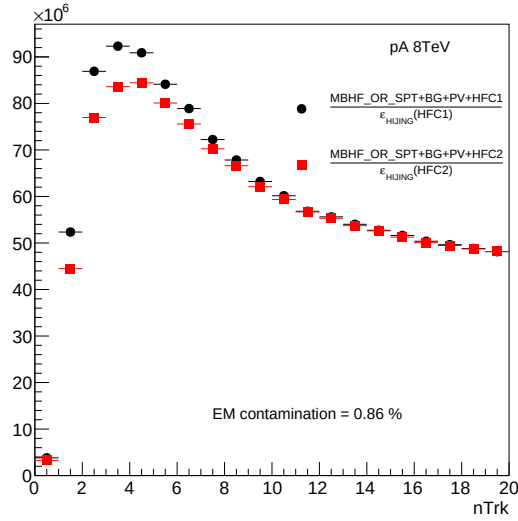


Figure 3: Estimation of remaining EM contamination after event selection with the HF coincidence filter with one tower per side.

3.4 Rejection of pileup events

For pPb 8 TeV data samples, the pileup scenario (pileup probability) is around 10-25% for the nominal runs depending on the fills and instantaneous luminosity, where the pileup is the number of interactions per bunch crossing. For this analysis, the pileup is a secondary effect, since all the tracks that used in calculating the observables are reconstructing with respect to one vertex. However, in order to make precise measurement, the effect from pileup is also evaluated and a proper filter has been applied to all the events. This pileup filter is not intended to remove events with two collisions, but to only reject the events that are too close to each other spatially. The details of the pileup filter are given below, while systematic checks have been done with different pileup filters and with different pileup samples in the data.

3.4.1 Samples for pileup study

For pileup study, we have used a few different samples to determine what cut values should be used.

For MC:

- pPb EPOS MinBias sample with exactly pileup = 1
- pPb EPOS MinBias sample with exactly pileup = 2.
- pPb EPOS MinBias sample with exactly pileup = 3.

For Data:

- pPb 8 TeV run = 285832, pileup 0.004
- pPb 8 TeV run = 285480, pileup 0.04
- pPb 8 TeV run = 285505, pileup 0.25
- pPb 8 TeV run = 285517, pileup 0.1.

3.4.2 Pileup filters

Multiple collisions events can be identified by the number of good primary vertices in the event. However, not all multiple collision events would lead to multiple reconstructed vertex, nor single collision event would lead to only one reconstructed vertex. In the multiple collision events with only one vertex reconstructed, the two collisions could be close enough that vertex reconstruction algorithm fails. In the case of a single collisions, the decay of a short lived particle close to the beam line may result in the reconstruction of an additional primary vertex (vertex splitting).

In order to balance between statistics and pileup contamination, one needs to look at the spatial distribution of the vertices, the correlation between number of tracks associated with the second populated vertex and dz value, to come up with a reasonable combinations for the pileup filter. Below a few cuts are proposed for pileup filters.

- dz cut : accept those events that have two vertices where dz , the z distance between the most populated vertex and other vertices, is larger than some value, i.e, 1 cm.
- Gplus cut: according to the number of tracks w.r.t the second populated vertex, cut on a dz value.
- vtx1 cut: accept events that only has one vertex.

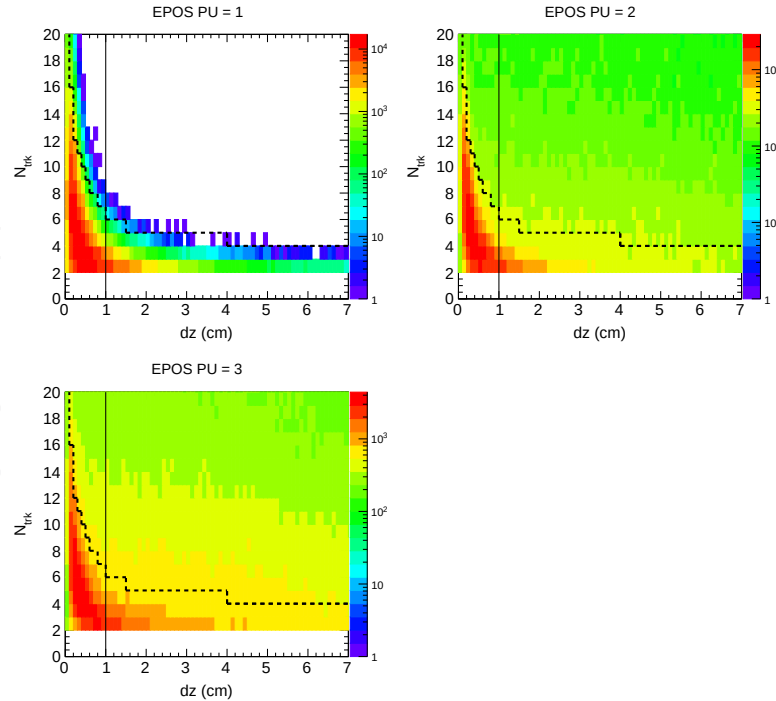
The available filters with instructions can be found in here, github, as,

Table 4: Table of N_{trk} vs dz cuts.

N_{trk}	dz
0–4	NA
5	4.0
6	1.5
7	1.0
8	0.6
9	0.5
10	0.4
11	0.3
12–15	0.2
16–19	0.1
20+	0.0

- *pileupFilter_pPb8TeV_vtx1*, with only 1 reconstructed vertex.
- *pileupFilter_pPb8TeV_Gplus*, with only N_{trk} vs dz cut.
- *olvFilter_pPb8TeV_dz1p0*, with N_{trk} vs dz cut plus dz cut at 1 cm. olvFilter stands for “overlapping vertex filter”, which only filters those events with 2 or more vertices are too close to each other.

In Table. 4, it shows the cut values in N_{trk} vs dz cut. From Fig. 4 to Fig. 5, they are showing the N_{trk} vs dz distribution in both MC and data to indicate where the cut values are.

Figure 4: N_{trk} vs dz from MC is shown and the line is indicating where the cut values are.

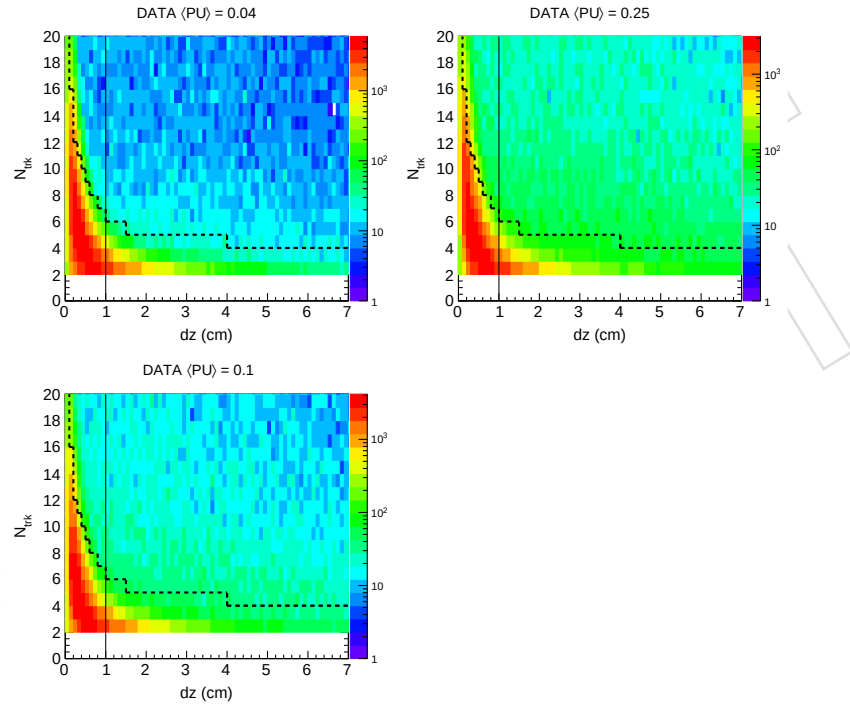


Figure 5: N_{trk} vs dz from Data is shown and the line is indicating where the cut values are.

3.4.3 Pileup Filter Performance

The available filters are applied on various MC samples and Data samples to see the filter efficiency. From Fig. 6 to Fig. 7, they show the accepted fraction of events after applying various filters as a function of $N_{\text{trk}}^{\text{offline}}$. The filter efficiency can be used as a measure of how much statistics would be reduced after using each filter as well as an estimate of the remaining pileup contaminations given the average pileup μ . The overall filter efficiency for *pileupFilter_pPb8TeV_Gplus* filter is around 99.6%, while the purity of the remaining sample is about 97.3%. In this analysis, the *pileupFilter_pPb8TeV_Gplus* filter is only applied for systematic studies and the default filter is *olvFilter_pPb8TeV_dz1p0*.

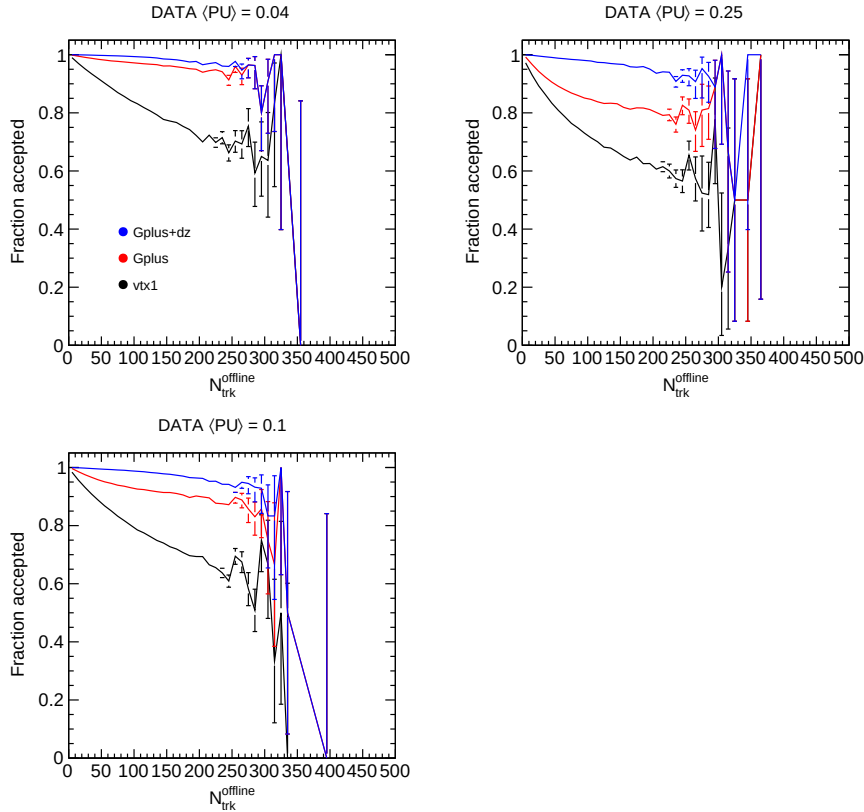


Figure 6: The accepted fraction of number of events is shown as a function $N_{\text{trk}}^{\text{offline}}$ with different defined pileup filters for different runs.

In order to make sure the *pileupFilter_pPb8TeV_Gplus* filter is effective in terms of removing pileup events, the HF sum E_T and $N_{\text{trk}}^{\text{offline}}$ are compared with different pileup samples before and after the filter. The distributions from different pileup samples should be expected to be the same if the pileup events are removed. In Fig. 8 and Fig. 9, the HF sum E_T and $N_{\text{trk}}^{\text{offline}}$ are shown before pileup filter, and clearly the HF sum E_T has a clear ordering with the level of pileup in the sample. For $N_{\text{trk}}^{\text{offline}}$, since our multiplicity definition is with respect to one vertex, the pileup effect is a second order effect that the difference between different pileup level is much smaller comparing to that of HF sum E_T . In Fig. 10 and Fig. 11, the same two distributions are shown after the pileup filter, and as expected, both distributions are almost the same showing the pileup events are mostly removed.

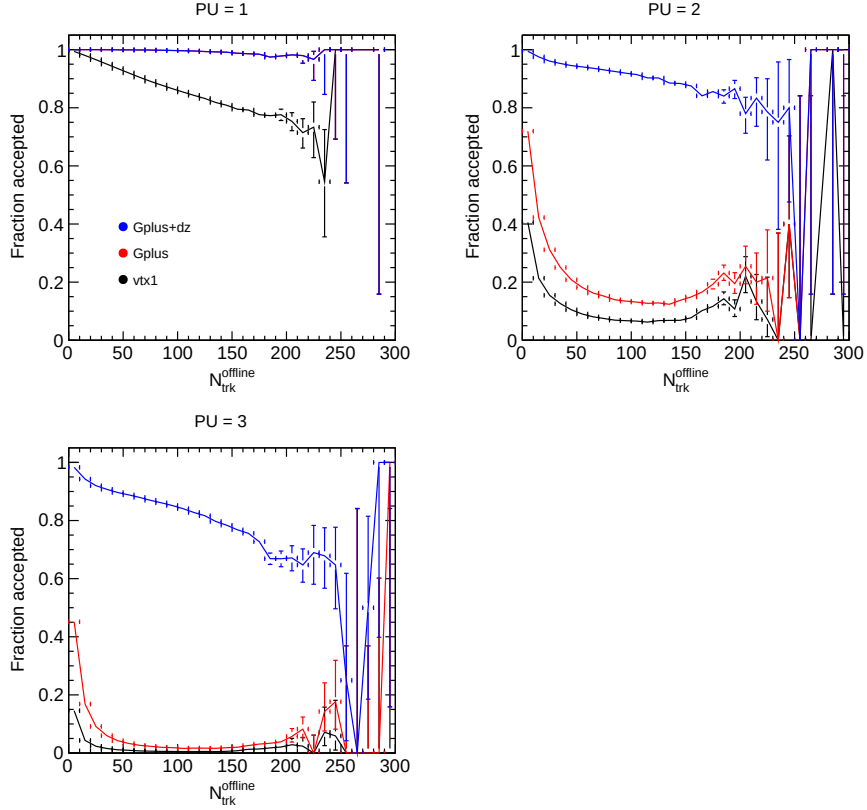


Figure 7: The accepted fraction of number of events is shown as a function $N_{\text{trk}}^{\text{offline}}$ with different defined pileup filters for different pileup samples.

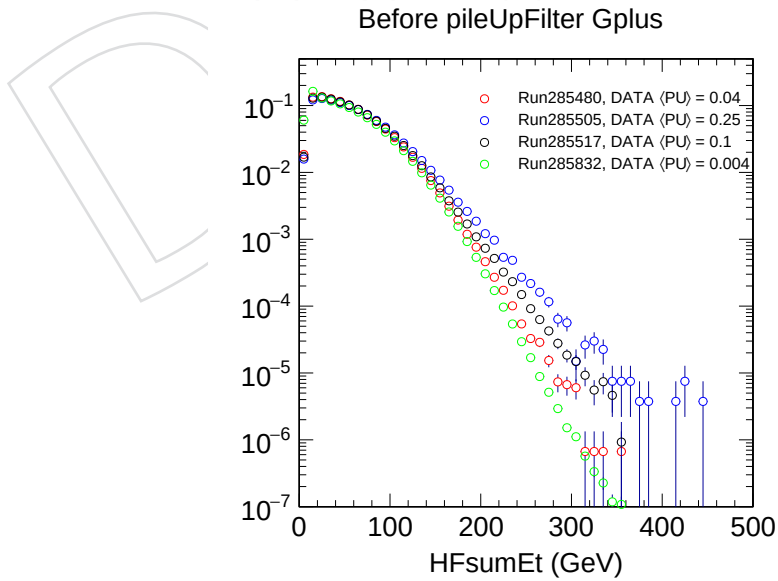


Figure 8: The HF sum E_T distributions with different pileup runs before the *pileupFilter_pPb8TeV_Gplus* are shown.

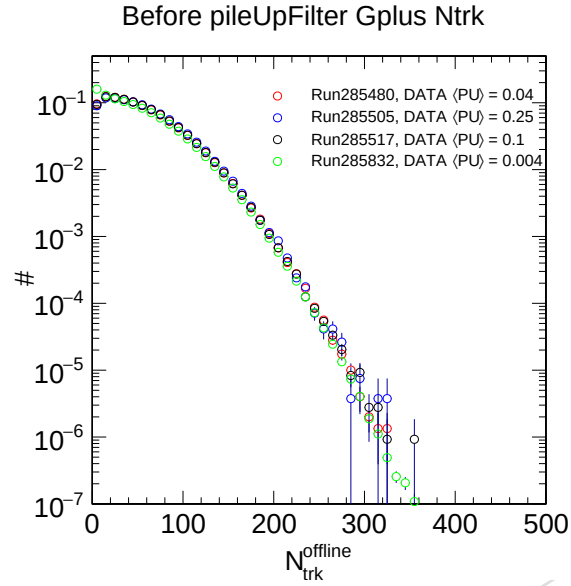


Figure 9: The $N_{\text{trk}}^{\text{offline}}$ distributions with different pileup runs before the *pileupFilter_pPb8TeV_Gplus* are shown.

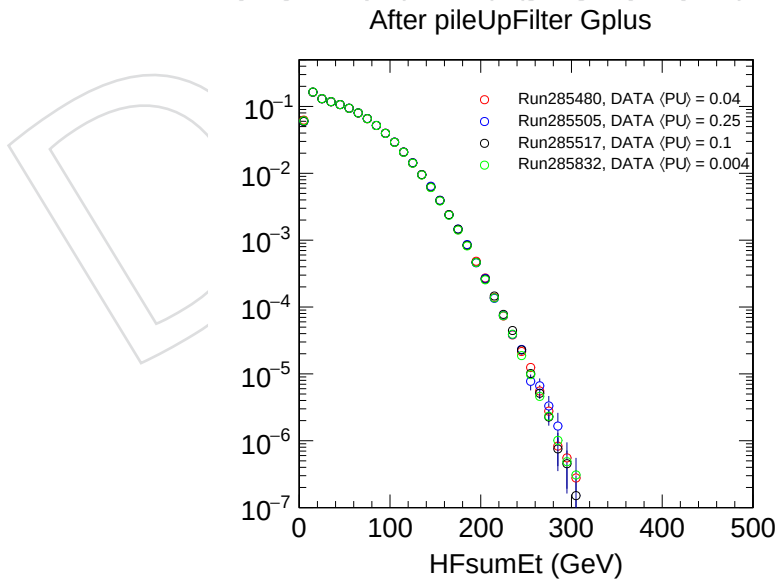


Figure 10: The HF sum E_T distributions with different pileup runs after the *pileupFilter_pPb8TeV_Gplus* are shown.

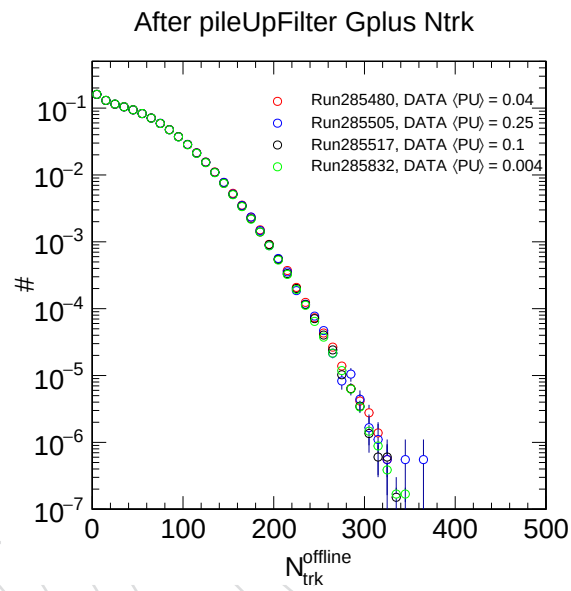


Figure 11: The $N_{\text{trk}}^{\text{offline}}$ distributions with different pileup runs after the *pileupFilter_pPb8TeV_Gplus* are shown.

The vertex merging rate, the probability of having 2 real collisions but with only one vertex reconstructed, needs to be estimated in order to understand the efficiency of the filters. For example, for the *pileUpFilter_pPb8TeV_vtx1*, we have filtered out all the events that have 2 reconstructed vertices, but the remaining events might still have 2 real collisions. In EPOS pileup = 2 samples, there are 10% chances that the algorithm fails to reconstruct a valid vertex, given there are two real collisions in the event. With the possible highest pileup scenario in the data (25%), the probability of having at least 2 real collisions in a MinBias sample, is around 11%, assuming the pileup follows Poisson statistics. Therefore, the vertex merging probability after pileup filters is around 1.1%.

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4 Muon selections and performance

The muon reconstruction algorithm starts either by finding tracks in the muon detectors, which are then fitted together with tracks reconstructed in the silicon tracker (*global muons*) or by extrapolating tracks from the silicon tracker to match a hit on at least one segment of the muon detectors (*tracker muons*). Muons in the kinematic range $(\mu) > 3 \text{ GeV}/c$, $|\eta(\mu)| < 2.4$ are selected, which allows J/ψ mesons to be well measured within the kinematic range of $1.4 < |y_{J/\psi}| < 2.4$ and $p_T > 0.2 \text{ GeV}/c$.

The muon candidates are further required to be particle flow muon candidates [42] and pass the *Soft Muon* ID selection, as endorsed by the Muon Physics Object Group (POG) for Run 2 analyses, which can be found here: (https://twiki.cern.ch/twiki/bin/view/CMS/SWGuideMuonIdRun2#Soft_Muon). The complete list of all the cuts applied on muons are as follows:

- muons are reconstructed as *particle flow muon candidate*;
- soft muon ID criteria:
 - TMOneStationTight (requires at least one well matched muon segment in any muon station for the track detected in the tracker);
 - the number of tracker layers with hits must be > 5 , to ensure sufficient transverse momentum resolution of muons;
 - the number of pixel layers with valid hits > 0 , to suppress muons from decays in flight;
 - the track high purity flag is used to reject bad quality tracks, which consists of a few outliers;
 - the distance between the primary event vertex and the muon track in the transverse plane, $D_{xy} < 0.3 \text{ cm}$, and the longitudinal plane, $D_z < 20.0 \text{ cm}$.

Combined acceptance and efficiency of single reconstructed muons with the above selections are studied using PYTHIA generator containing a J/ψ meson decaying into $\mu^+\mu^-$. A gen-level muon particle is considered to be reconstructed if it is matched to a reconstructed-level muon (with the above selections) within an angular cone size of $\Delta R < 0.03$. The combined acceptance and efficiency in 2D as a function of muon η and p_T , and in 1D as a function of muon p_T averaged over $1.4 < |\eta| < 2.4$ are shown in Fig. 12. The combined acceptance and efficiency in 2D as a function of muon η and total momentum is also shown in Fig. 13.

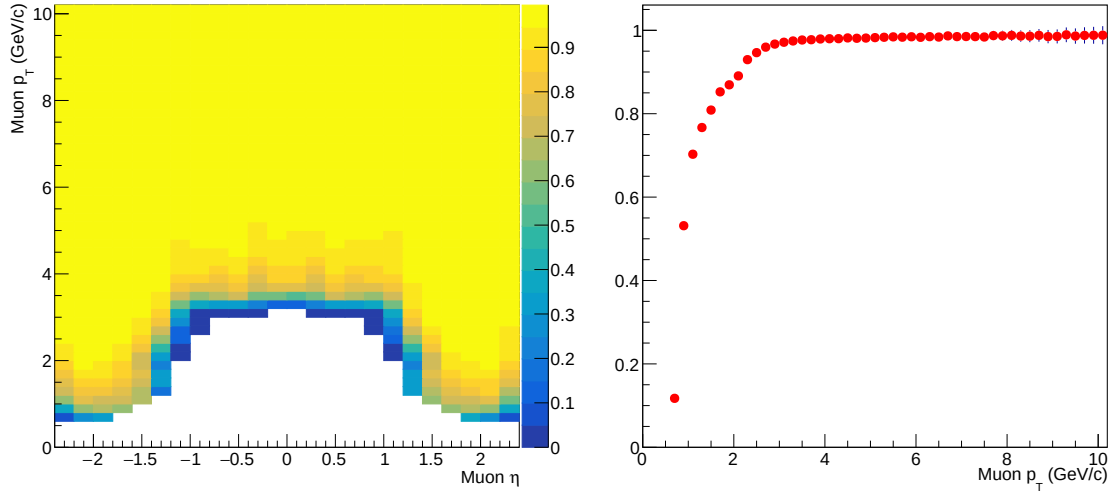


Figure 12: The combined acceptance and efficiency for single muons in 2D as a function of muon η and p_T (left), and in 1D as a function of muon p_T averaged over $1.4 < |\eta| < 2.4$ (right).

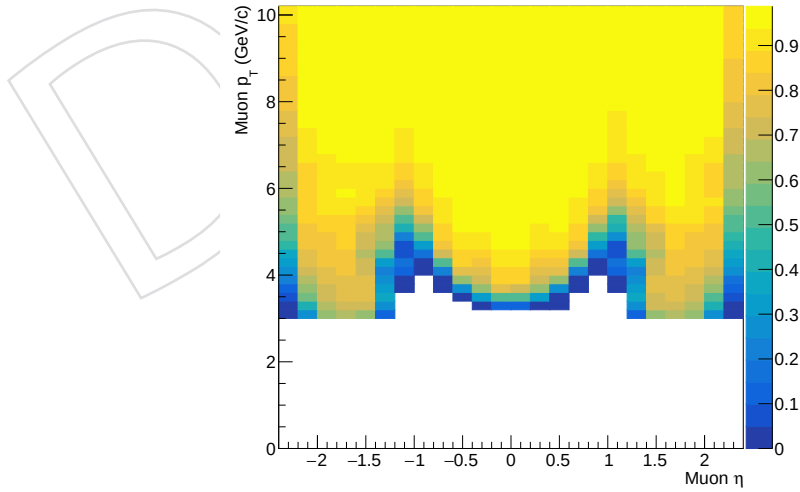


Figure 13: The combined acceptance and efficiency for single muons in 2D as a function of muon η and total momentum.

5 Reconstruction of J/ψ candidates

J/ψ candidates are reconstructed by combining pairs of oppositely charged PF muon candidates, fulfilling the soft muon ID selections as described in Sec. 4, within an invariant mass window within $0.6 \text{ GeV}/c^2$ from the nominal J/ψ mass from particle data group (PDG). The secondary (decay) vertex is reconstructed by using a ROOT "KinematicParticleVertexFitter", where least-mean squared minimization is implemented. In order to suppress combinatorial backgrounds, the vertex fit probability is required to be larger than 1%. The J/ψ kinematic range of $1.4 < |y| < 2.4$ and $0.2 < p_T < 10 \text{ GeV}/c$ is used in the correlation analysis later, to ensure good acceptance and efficiency

Combined acceptance and efficiency of reconstructed J/ψ mesons are studied using PYTHIA generator containing a J/ψ meson decaying into $\mu^+\mu^-$, by matching daughter muons at generator- and reconstructed-level, as shown in Fig. 14. An efficiency of 30–40% can be achieved at low p_T ($< 3 \text{ GeV}$) for forward region ($1.4 < |y| < 2.4$).

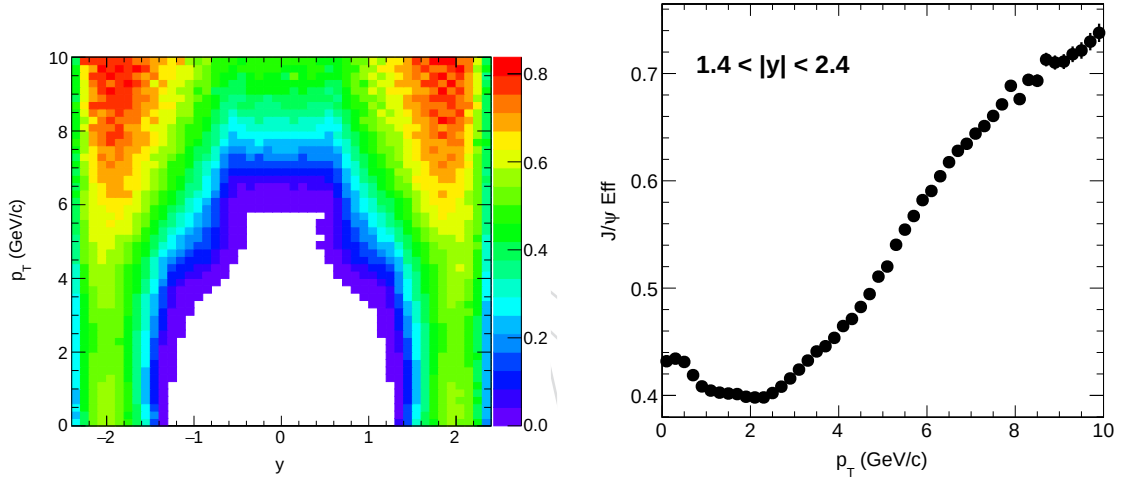


Figure 14: The combined acceptance and efficiency for J/ψ meson in 2D as a function of y and p_T (left), and in 1D as a function of p_T averaged over $1.4 < |y| < 2.4$ (right).

5.1 Signal extraction

In this analysis, a mass spectrum is always needed to get J/ψ v_2 , as described in Section 6.2. The fit procedure and model is studied in details in HIN-16-004 (AN-16-067). The fit function consists of the following components:

- Weighted sum of two Crystal Ball functions $g_{CB}(m_{\mu^+\mu^-})$ ($f \cdot g_{CB_1}(m_{\mu^+\mu^-}) + (1 - f) \cdot g_{CB_2}(m_{\mu^+\mu^-})$) to model the J/ψ signal. The two Crystal Ball functions have common mean m_0 and tail parameters α and n .
- 3rd order polynomial to model the combinatorial background.

The Crystal Ball function $g_{CB}(m)$ combines a Gaussian core and a power-law tail with an exponent n to account for energy loss due to final-state photon radiation, and a parameter α which defines the transition between the Gaussian and the power-law functions,

$$g_{CB}(m) = \begin{cases} \frac{N}{\sqrt{2\pi}\sigma_{CB}} \exp\left(-\frac{(m-m_0)^2}{2\sigma_{CB}^2}\right), & \text{for } \frac{m-m_0}{\sigma_{CB}} > -\alpha; \\ \frac{N}{\sqrt{2\pi}\sigma_{CB}} \left(\frac{n}{|\alpha|}\right)^n \exp\left(-\frac{|\alpha|^2}{2}\right) \left(\frac{n}{|\alpha|} - |\alpha| - \frac{m-m_0}{\sigma_{CB}}\right)^{-n}, & \text{for } \frac{m-m_0}{\sigma_{CB}} \leq -\alpha. \end{cases} \quad (1)$$

314 The Crystal Ball tail parameters obtained from the J/ψ MC simulations are shown in Tabs. 5.
 315 The tail parameters α and n are fixed to the values found in MC (p_T -centrality integrated fits)
 316 in data fits.

Table 5: Signal shape parameters obtained in J/ψ MC for the double Crystal Ball function.

p_T [GeV/c]	α	n
0–1.8	4.31	1.89
1.8–3.0	3.51	1.98
3.0–4.5	3.03	2.03
4.5–6.0	2.74	2.07
6.0–8.0	2.38	2.11
8.0–10.0	2.11	2.15

317 Figure 15 shows mass spectrum fit for pPb collisions with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$.

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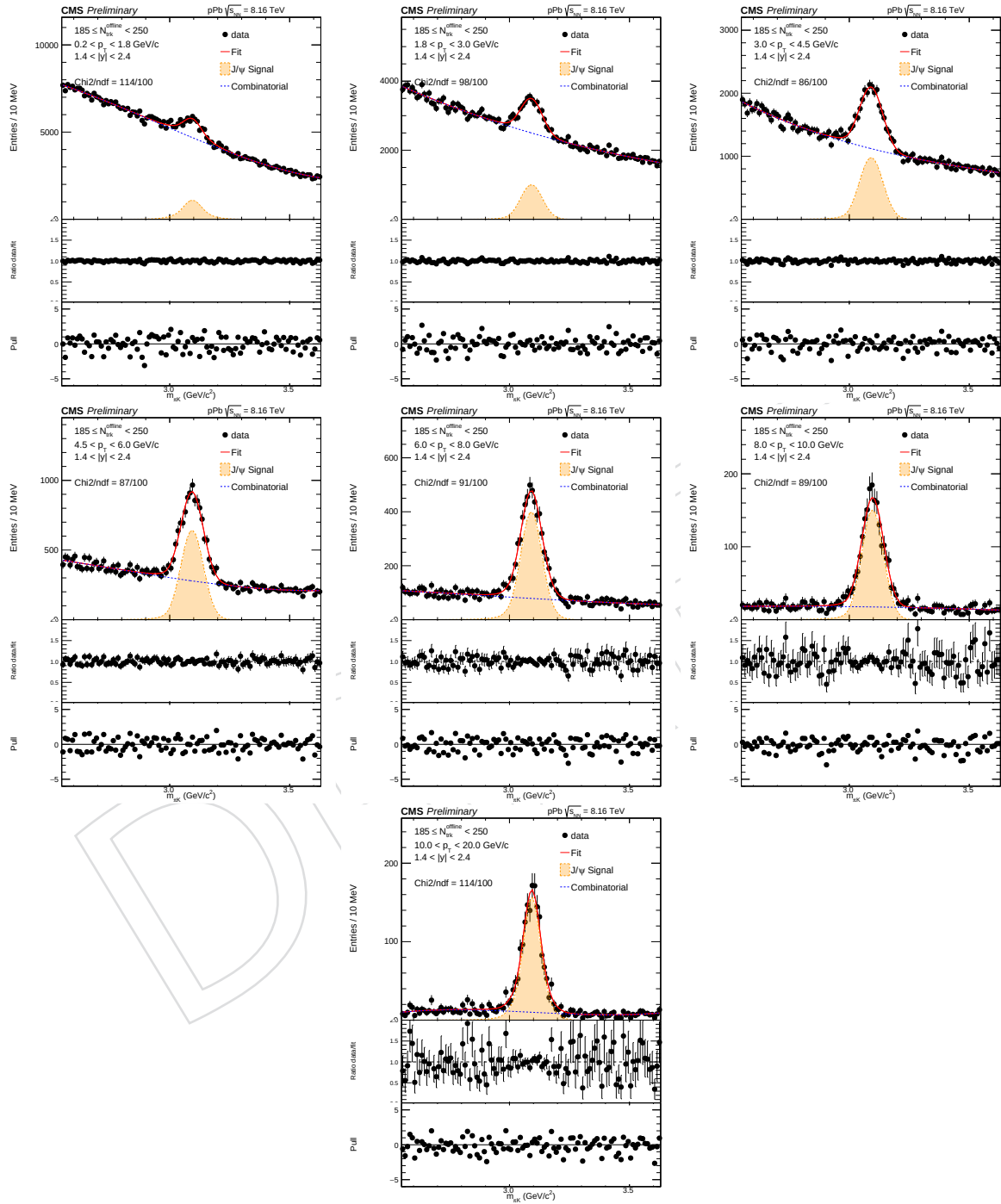


Figure 15: Mass spectrum fit in different p_T bins for pPb collisions with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$.

5.2 Non-Prompt J/ψ rejection

The procedure of rejecting non-prompt J/ψ has been studied in detail in HIN-16-004 (AN-16-067). Selected information are provided in this section.

5.2.1 The pseudo-proper decay length $\ell_{J/\psi}^{3D}$

The J/ψ mesons coming from b-hadron decays (non-prompt charmonia) are identified by the measurement of a secondary $\mu^+\mu^-$ vertex displaced from the primary collision vertex. The displacement vector between the $\mu^+\mu^-$ vertex and the primary vertex is measured taking into account all directions: x, y and z. The non-prompt charmonia are separated by cutting on pseudo-proper decay length,

$$\ell_{J/\psi}^{3D} = L_{xyz} \cdot m / p, \quad (2)$$

where m is the PDG J/ψ mass, assumed for all dimuons. This quantity $\ell_{J/\psi}^{3D}$ is computed from L_{xyz} , where L_{xyz} is the most probable b-hadron decay length in the laboratory frame [43, 44].

In this analysis, we measure prompt J/ψ v_2 by rejecting dimuons with $\ell_{J/\psi}^{3D}$ above a certain value.

5.2.2 Definition of the cut on the pseudo-proper b-hadron decay length

The $\ell_{J/\psi}^{3D}$ cut is tuned based on the efficiency of keeping (rejecting) prompt (non-prompt) J/ψ as performed in HIN-16-004 (AN-16-067), where the efficiency is meant for the fraction of events passing a certain $\ell_{J/\psi}^{3D}$ cut. The strategy is to remove as many non-prompt J/ψ as possible, while keeping the efficiency for prompt J/ψ high. The cut is chosen for each bin in the analysis in such a way that the efficiency for prompt J/ψ is fixed at 90%, which corresponds to a rejection of non-prompt J/ψ of 70–90% or more, as shown in Sec. 5.2.3.

The distributions of $\ell_{J/\psi}^{3D}$ and an illustration of how we determine the cut values are shown in Fig. 16 for $3 < p_T < 4.5$ GeV/c. Since the cut values vary with the J/ψ p_T , we calculated them in different p_T bins and fitted the obtained pattern by the function $\ell_{J/\psi}^{3D}(p_T) = a + b/p_T + c * p_T + d * p_T^2$. Figure. 17 shows the dependence of the cut values on p_T and the fitted function used for the parametrisation. It is important to note that uncertainties on this parametrisation are irrelevant, and are therefore not given. Then in the analysis the prompt J/ψ are obtained cutting on $\ell_{J/\psi}^{3D}$, which has to be smaller than the value estimated from the J/ψ p_T from the aforementioned $a + b/p_T + c * p_T + d * p_T^2$ function.

Table. 6 provides the fraction of J/ψ signal (obtained from fits to the mass spectrum) passing the $\ell_{J/\psi}^{3D}$ cuts in pPb data for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ and $N_{\text{trk}}^{\text{offline}} < 35$.

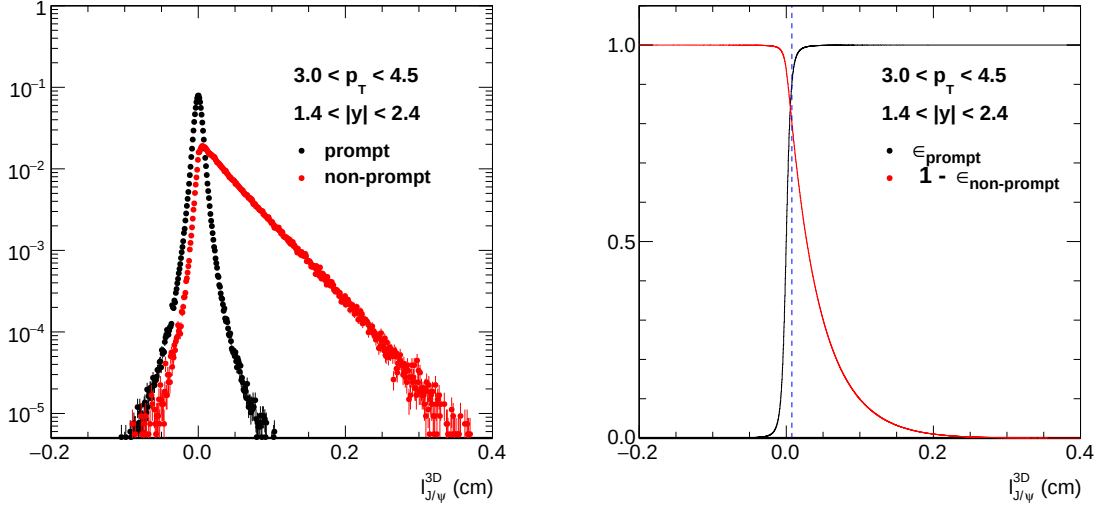


Figure 16: Pseudo-proper decay length distributions of prompt and non-prompt J/ψ in pPb MC simulations, for $3 < p_T < 4.5$ GeV/c and $1.4 < |y| < 2.4$. The right panel shows the prompt efficiency and the non-prompt rejection efficiency, for the chosen $\ell_{J/\psi}^{3D}$ cut.

Table 6: The fraction of J/ψ signal (obtained from fits to the mass spectrum) passing the $\ell_{J/\psi}^{3D}$ cuts in pPb data.

p_T range (GeV/c)	passing fraction, $185 \leq N_{\text{trk}}^{\text{offline}} < 250$	passing fraction, $N_{\text{trk}}^{\text{offline}} < 35$
0.2–1.8	85.2%	82.4%
1.8–3.0	79.0%	78.2%
3.0–4.5	75.2%	75.7%
4.5–6.0	76.8%	74.9%
6.0–8.0	74.8%	71.6%
8.0–10.0	71.0%	67.8%

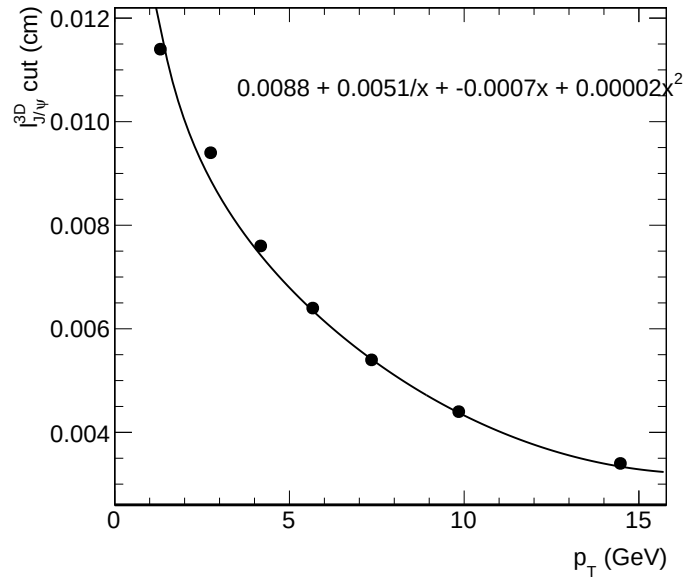


Figure 17: The dependence with p_T of the $\ell_{J/\psi}^{3D}$ cut value giving a 90% efficiency for prompt J/ ψ simulation, for $1.4 < |y| < 2.4$. The cut values are fitted by a function ($a + b/p_T + c * p_T + d * p_T^2$). Fit parameters are also shown.

5.2.3 $\ell_{J/\psi}^{3D}$ cut efficiency

The efficiencies of the $\ell_{J/\psi}^{3D}$ cut alone as a function of p_T , can be found in Fig. 18 for MC. By construction, the efficiencies are around 90% for prompt J/ψ . The efficiency for non-prompt J/ψ is found to be up to around 30% at low p_T .

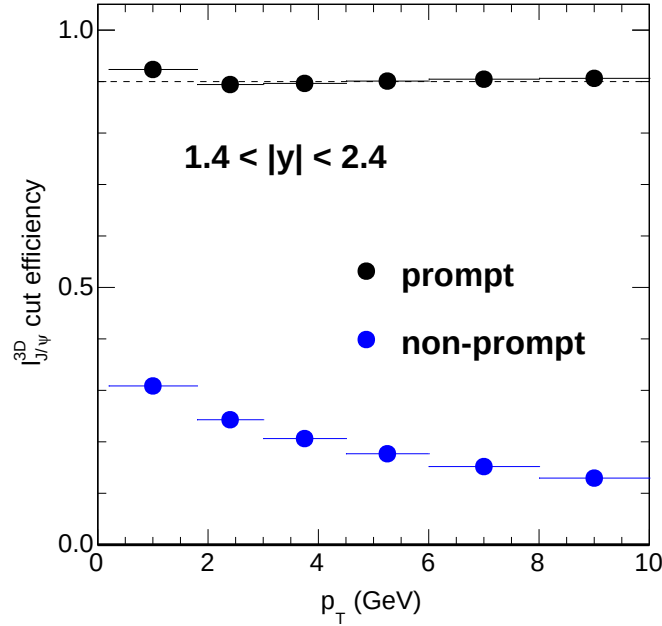


Figure 18: Efficiency of the $\ell_{J/\psi}^{3D}$ cut as a function of p_T in MC, for $1.4 < |y| < 2.4$. A dotted line at 90% has been added to guide the eye.

5.2.4 Comparison of $\ell_{J/\psi}^{3D}$ distribution for J/ψ between data and prompt MC

In order to check the agreement between data and MC for the $\ell_{J/\psi}^{3D}$ distribution and resolution, the $\ell_{J/\psi}^{3D}$ distribution in pPb data and in prompt J/ψ MC are compared in Fig. 19 for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ and in Fig. 20 for $N_{\text{trk}}^{\text{offline}} < 35$. For data, candidates from combinatorial background are not expected to follow the MC $\ell_{J/\psi}^{3D}$ distribution. Therefore, better agreement has been observed between data and MC for $N_{\text{trk}}^{\text{offline}} < 35$ than for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$, and for high p_T than for low p_T .

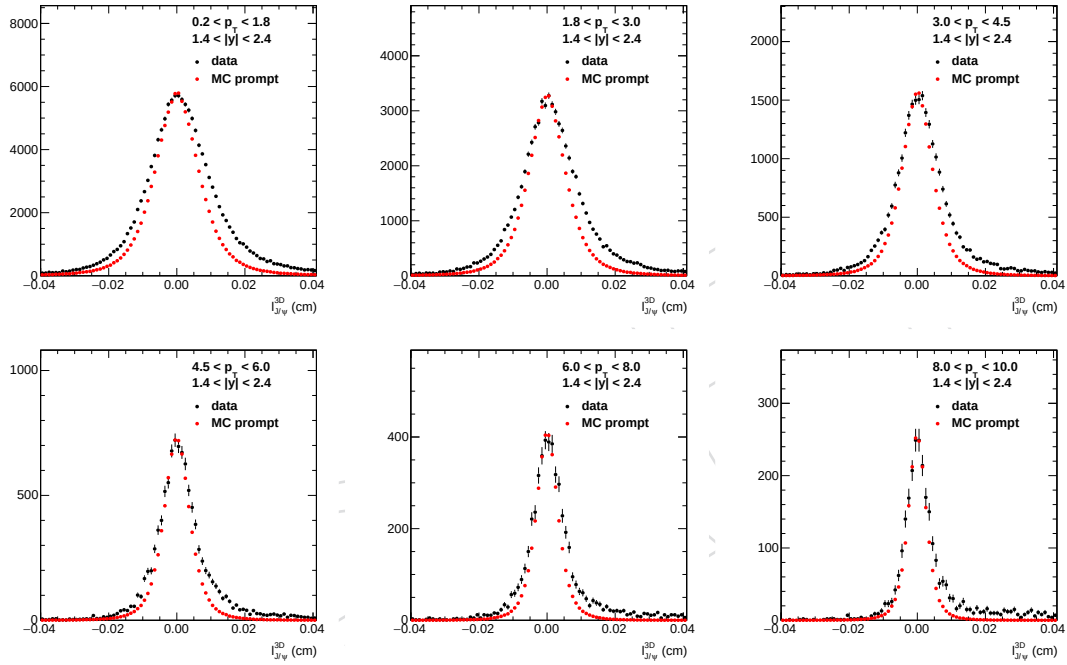


Figure 19: Comparison of the $\ell_{J/\psi}^{3D}$ distribution in pPb data (black) and prompt J/ψ MC (red), for different p_T bins for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. The MC distribution is normalized to the data in the range of $-0.002 < \ell_{J/\psi}^{3D} < 0.002$ cm. All analysis cuts are applied except the $\ell_{J/\psi}^{3D}$ cut. The J/ψ mass is required to be between 2.9 and 3.25 GeV.

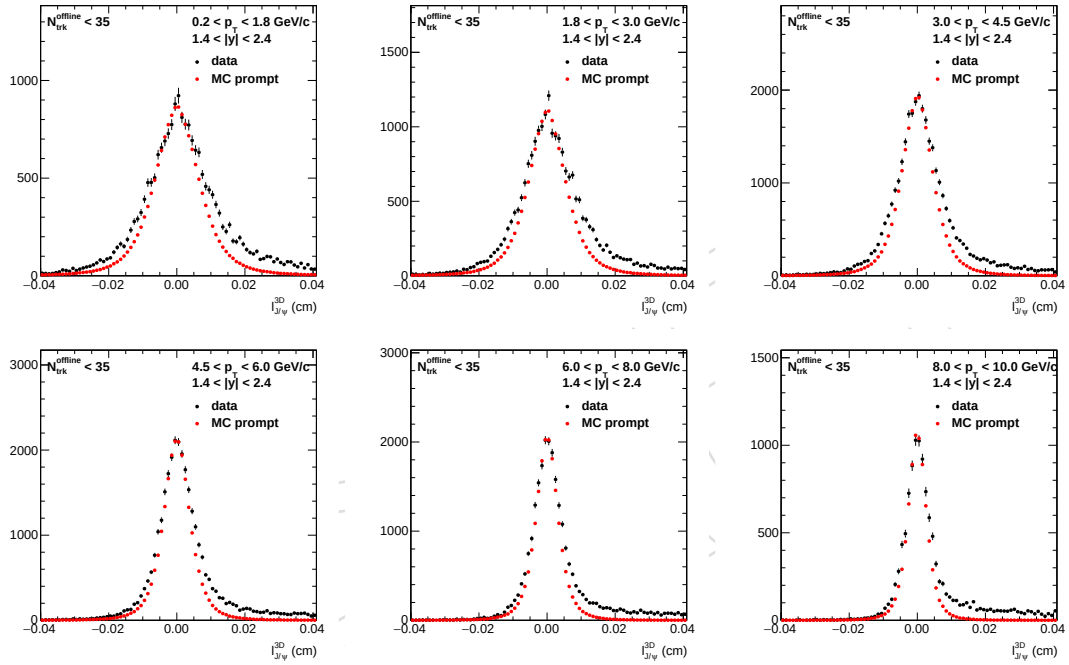


Figure 20: Comparison of the $\ell_{J/\psi}^{3D}$ distribution in pPb data (black) and prompt J/ψ MC (red), for different p_T bins for $N_{\text{trk}}^{\text{offline}} < 35$. The MC distribution is normalized to the data in the range of $-0.002 < \ell_{J/\psi}^{3D} < 0.002$ cm. All analysis cuts are applied except the $\ell_{J/\psi}^{3D}$ cut. The J/ψ mass is required to be between 2.9 and 3.25 GeV.

5.2.5 Unfolding $\ell_{J/\psi}^{3D}$ distribution for signal J/ψ

In order to get rid of effect from the background candidates in the comparison of $\ell_{J/\psi}^{3D}$ between data and MC, data $\ell_{J/\psi}^{3D}$ distributions have been unfolded to get the $\ell_{J/\psi}^{3D}$ for J/ψ signal using RooStats::SPlot. The $\ell_{J/\psi}^{3D}$ distribution in pPb data for signal J/ψ and in prompt J/ψ MC are compared in Fig. 21 for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ and in Fig. 22 for $N_{\text{trk}}^{\text{offline}} < 35$. A good agreement has been found for $\ell_{J/\psi}^{3D}$ distributions in data and in MC for prompt J/ψ, while there are contributions from non-prompt J/ψ in data.

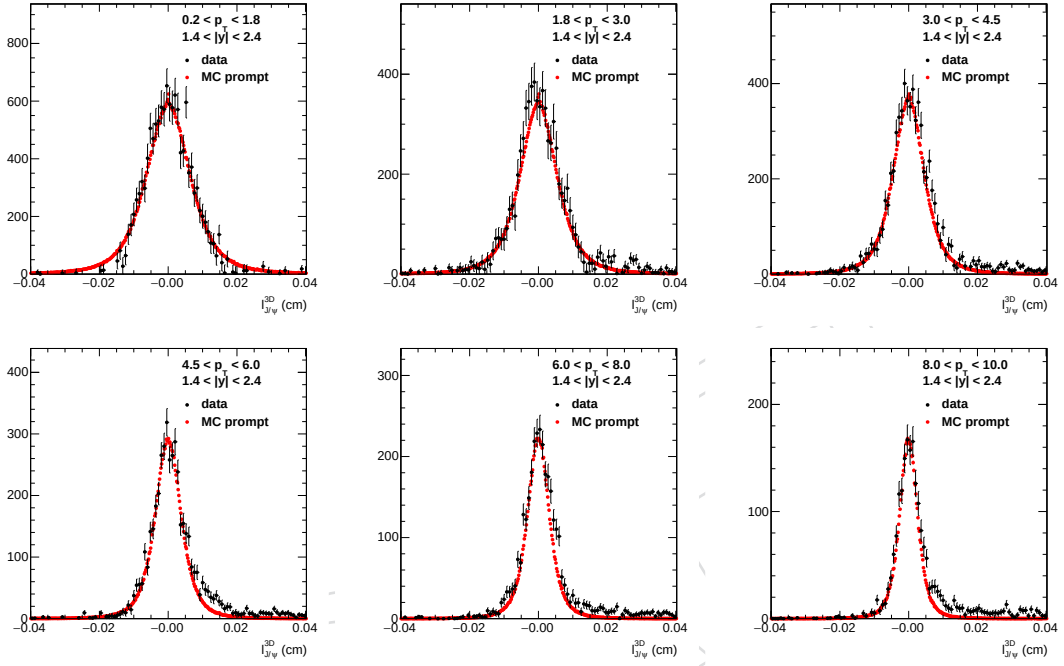


Figure 21: Comparison of the $\ell_{J/\psi}^{3D}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and prompt J/ψ MC (red), for different p_T bins for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. The MC distribution is normalized to the data in the range of $-0.004 < \ell_{J/\psi}^{3D} < 0.004$ cm. All analysis cuts are applied except the $\ell_{J/\psi}^{3D}$ cut.

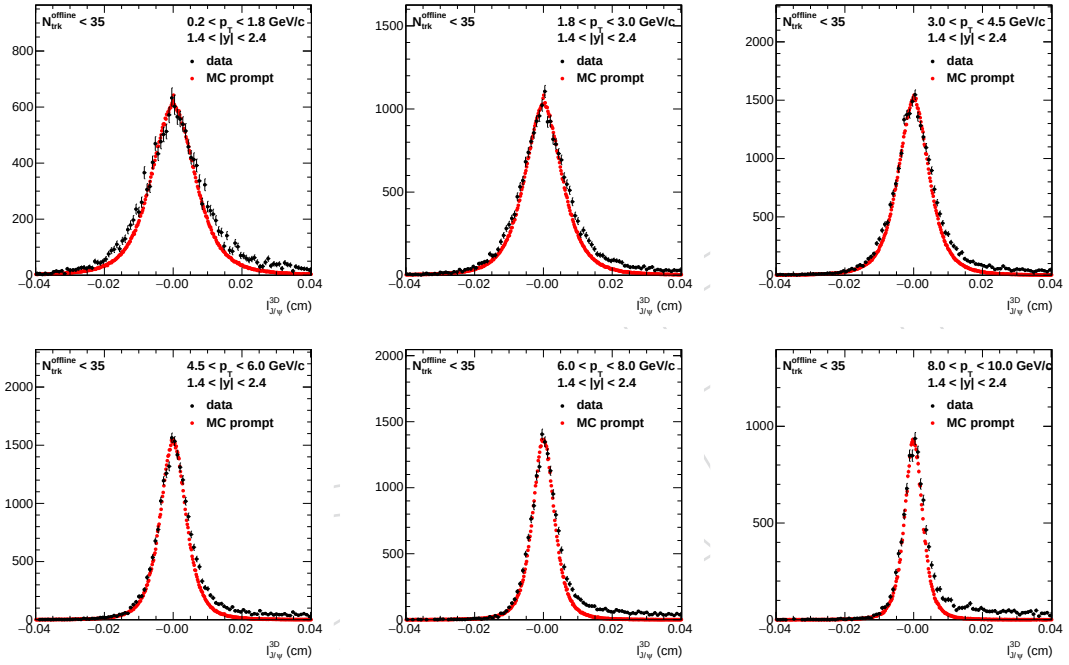


Figure 22: Comparison of the $\ell_{J/\psi}^{3D}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and prompt J/ψ MC (red), for different p_T bins for $N_{trk}^{offline} < 35$. The MC distribution is normalized to the data in the range of $-0.004 < \ell_{J/\psi}^{3D} < 0.004$ cm. All analysis cuts are applied except the $\ell_{J/\psi}^{3D}$ cut.

5.2.6 Imbalance of $\ell_{J/\psi}^{3D}$ distribution for non-prompt J/ψ in data and MC

In order to check if the non-prompt J/ψ $\ell_{J/\psi}^{3D}$ distributions in data and MC are the same, i.e. the non-prompt rejection efficiency is same in data and MC, a check on the imbalance of $\ell_{J/\psi}^{3D}$ distribution for non-prompt J/ψ has been performed. The negative part ($\ell_{J/\psi}^{3D} < 0$) of the $\ell_{J/\psi}^{3D}$ distributions in Fig. 21 has been reflected with respect to $\ell_{J/\psi}^{3D} = 0$, and then subtracted from the positive part of the $\ell_{J/\psi}^{3D}$ distributions. Since the prompt J/ψ $\ell_{J/\psi}^{3D}$ distributions are symmetric, after the subtraction there is no contribution from prompt J/ψ left. The result distributions represent the imbalance of $\ell_{J/\psi}^{3D}$ distributions for non-prompt J/ψ . The comparison of those distributions can reveal possible difference between non-prompt J/ψ $\ell_{J/\psi}^{3D}$ distributions in data and MC. Fig. 23 and 24 show the comparison for all p_T bins, where within uncertainties no difference has been observed between data and MC. Furthermore, Fig. 25 shows the $\ell_{J/\psi}^{3D}$ cut efficiency from the $\ell_{J/\psi}^{3D}$ imbalance distributions, which are identical within uncertainties. Therefore, the $\ell_{J/\psi}^{3D}$ distributions for non-prompt J/ψ is expected to be same in data and MC.

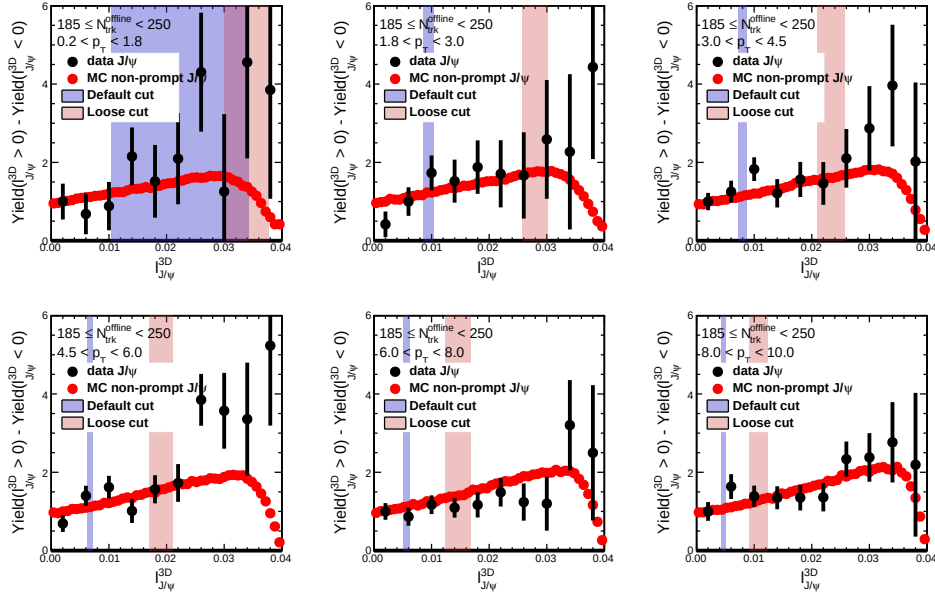


Figure 23: Comparison of the imbalance in $\ell_{J/\psi}^{3D}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (red), for different p_T bins for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. All analysis cuts are applied except the $\ell_{J/\psi}^{3D}$ cut. The $\ell_{J/\psi}^{3D}$ cuts are indicated by color bands.

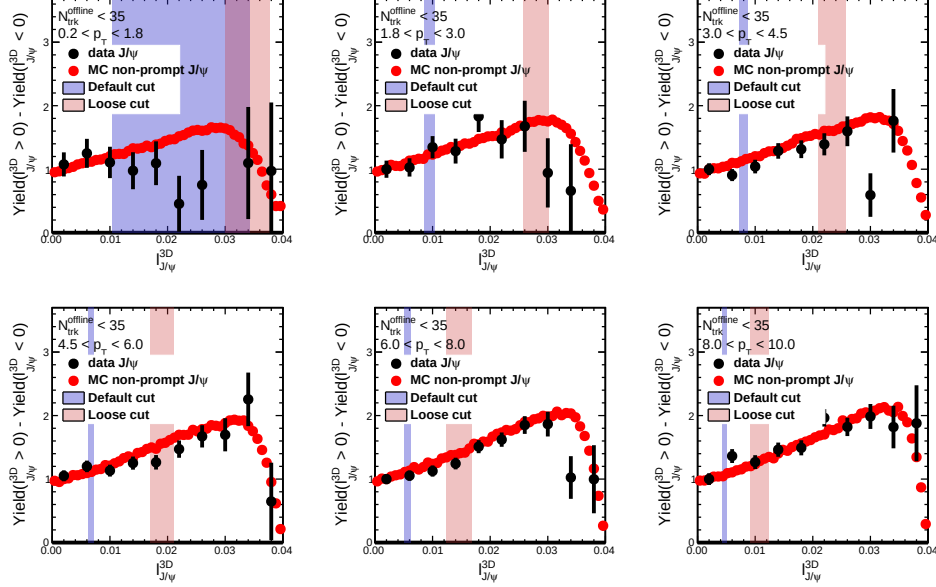


Figure 24: Comparison of the imbalance in $\ell_{J/\psi}^{3D}$ distribution in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (red), for different p_T bins for $N_{\text{trk}}^{\text{offline}} < 35$. All analysis cuts are applied expect the $\ell_{J/\psi}^{3D}$ cut. The $\ell_{J/\psi}^{3D}$ cuts are indicated by color bands.

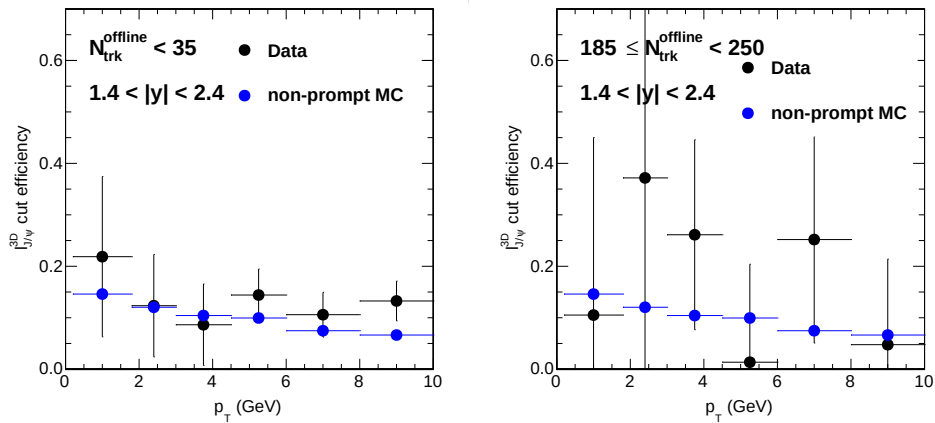


Figure 25: $\ell_{J/\psi}^{3D}$ cut efficiency from the $\ell_{J/\psi}^{3D}$ imbalance distributions in pPb data for signal J/ψ from SPlot unfolding (black) and non-prompt J/ψ MC (blue) for $N_{\text{trk}}^{\text{offline}} < 35$ (left) and $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ (right).

5.2.7 Estimation of non-prompt J/ψ fraction in data before and after $\ell_{J/\psi}^{3D}$ cut

It has been shown in Sec. 5.2.5 and 5.2.6 that $\ell_{J/\psi}^{3D}$ distributions for both prompt and non-prompt J/ψ have good agreement in data and MC, i.e. $\ell_{J/\psi}^{3D}$ cut efficiency in data and MC are identical. Using the $\ell_{J/\psi}^{3D}$ cut efficiency and the fraction of J/ψ passing the $\ell_{J/\psi}^{3D}$ cut in Table. 6, an estimation of non-prompt J/ψ fraction in data is possible.

The fraction of J/ψ passing the $\ell_{J/\psi}^{3D}$ cut (defined as eff_{cut}) can be expressed as combination of prompt and non-prompt J/ψ passing the $\ell_{J/\psi}^{3D}$ cut,

$$eff_{cut} = f_{prompt} * eff_{prompt} + (1 - f_{prompt}) * eff_{non-prompt}, \quad (3)$$

where f_{prompt} is the prompt fraction (before $\ell_{J/\psi}^{3D}$ cut is applied) in data and eff_{prompt} ($eff_{non-prompt}$) is the $\ell_{J/\psi}^{3D}$ cut efficiency for prompt (non-prompt) J/ψ in Fig. 14. Table. 7 provides the calculated f_{prompt} for each p_T bin without $\ell_{J/\psi}^{3D}$ cut. Furthermore, prompt fraction can be easily estimated for J/ψ passing $\ell_{J/\psi}^{3D}$ cut, shown in Table. 8.

Table 7: The prompt J/ψ fraction f_{prompt} in pPb data without $\ell_{J/\psi}^{3D}$ cut.

p_T range (GeV/c)	$f_{prompt}, 185 \leq N_{trk}^{offline} < 250$	$f_{prompt}, N_{trk}^{offline} < 35$
0.2–1.8	88.4%	83.8%
1.8–3.0	84.0%	82.8%
3.0–4.5	79.1%	79.8%
4.5–6.0	81.7%	79.0%
6.0–8.0	79.2%	74.9%
8.0–10.0	74.7%	70.6%

Table 8: The prompt J/ψ fraction f_{prompt} in pPb data with $\ell_{J/\psi}^{3D}$ cut.

p_T range (GeV/c)	$f_{prompt}, 185 \leq N_{trk}^{offline} < 250$	$f_{prompt}, N_{trk}^{offline} < 35$
0.2–1.8	95.8%	93.9%
1.8–3.0	95.1%	94.7%
3.0–4.5	94.3%	94.5%
4.5–6.0	95.8%	95.0%
6.0–8.0	95.8%	94.7%
8.0–10.0	95.4%	94.4%

6 Analysis Technique

6.1 Dihadron correlation technique

The v_n harmonics for J/ψ particles are measured with a dihadron correlation technique follows exactly the same procedure as was established in Refs. [5, 6, 18], which was documented in detail in the previous analysis note, AN-12-352 (HIN-12-015). For each track multiplicity bin, trigger particles are defined as all J/ψ candidates with $1.4 < |y| < 2.4$ and in specified p_T^{trig} range. The number of trigger particles in the event is denoted by N_{trig} , and there may be more than one trigger particle per event. Hadron pairs are formed by associating with every trigger particle the remaining charged particles with $|\eta| < 2.4$ and in a specified p_T^{assoc} range. The per-trigger-particle associated yield distribution is then defined by:

$$\frac{1}{N_{\text{trig}}} \frac{d^2 N^{\text{pair}}}{d\Delta\eta d\Delta\phi} = B(0,0) \times \frac{S(\Delta\eta, \Delta\phi)}{B(\Delta\eta, \Delta\phi)}, \quad (4)$$

where $\Delta\eta$ and $\Delta\phi$ are the differences in η and ϕ of the pair, respectively. The signal distribution, $S(\Delta\eta, \Delta\phi)$, is the measured per-trigger-particle distribution of same-event pairs, i.e.,

$$S(\Delta\eta, \Delta\phi) = \frac{1}{N_{\text{trig}}} \frac{d^2 N^{\text{same}}}{d\Delta\eta d\Delta\phi}. \quad (5)$$

The mixed-event background distribution,

$$B(\Delta\eta, \Delta\phi) = \frac{1}{N_{\text{trig}}} \frac{d^2 N^{\text{mix}}}{d\Delta\eta d\Delta\phi}. \quad (6)$$

is constructed by pairing the trigger particles in each event with the associated particles from 20 different random events, excluding the original event. The 20 random events are not required to contain a J/ψ candidate. The symbol N^{mix} denotes the number of pairs resulting from the event mixing. The normalization of both signal and background distributions by dividing by N_{trig} is done after summing up the pair density distributions for all the events and then dividing by the total number of trigger particles from all the events.

The background distribution is used to account for random combinatorial background and pair-acceptance effects. The normalization factor $B(0,0)$ is the value of $B(\Delta\eta, \Delta\phi)$ at $\Delta\eta = 0$ and $\Delta\phi = 0$ (with a bin width of 0.3 in $\Delta\eta$ and $\pi/16$ in $\Delta\phi$), representing the mixed-event associated yield for both particles of the pair going in approximately the same direction, thus having full pair acceptance. Therefore, the ratio $B(0,0)/B(\Delta\eta, \Delta\phi)$ is the pair-acceptance correction factor used to derive the corrected per-trigger-particle associated yield distribution. Equation (4) is calculated in 2 cm wide bins of the vertex position (z_{vtx}) along the beam direction and averaged over the range $|z_{\text{vtx}}| < 15$ cm.

Each reconstructed track is weighted by the inverse of the efficiency factor, $\varepsilon_{\text{trk}}(\eta, p_T)$, as a function of the track's pseudorapidity and transverse momentum. The efficiency weighting factor accounts for the detector acceptance $A(\eta, p_T)$, the reconstruction efficiency $E(\eta, p_T)$, and the fraction of misidentified tracks, $F(\eta, p_T)$,

$$\varepsilon_{\text{trk}}(\eta, p_T) = \frac{AE}{1 - F}. \quad (7)$$

422 To further quantify the correlation structure, the 2-D distributions are reduced to one-dimensional
 423 (1-D) distributions in $\Delta\phi$ by averaging over the $\Delta\eta$ range [5, 6, 14, 18]. The azimuthal anisotropy
 424 harmonics are determined from a Fourier decomposition of long-range two-particle $\Delta\phi$ corre-
 425 lation functions,

$$\frac{1}{N_{\text{trig}}} \frac{dN^{\text{pair}}}{d\Delta\phi} = \frac{N_{\text{assoc}}}{2\pi} \left\{ 1 + \sum_n 2V_{n\Delta} \cos(n\Delta\phi) \right\}, \quad (8)$$

426 as described in Refs. [5, 6], where $V_{n\Delta}$ are the Fourier coefficients and N_{assoc} represents the total
 427 number of pairs per trigger particle for a given $(p_T^{\text{trig}}, p_T^{\text{assoc}})$ bin. The first three Fourier terms
 428 are included in the fits. Including additional terms have negligible effects on the fit results. A
 429 minimum η gap of 1 units is applied to remove short-range correlations from jet fragmentation.
 430 The elliptic anisotropy harmonics, $v_2\{2, |\Delta\eta| > 1\}$, of J/ψ from two-particle correlation method
 431 can be extracted from the fitted Fourier coefficients as a function of p_T ,

$$v_n(p_T^{\text{trig}}) = \frac{V_{n\Delta}(p_T^{\text{trig}}, p_T^{\text{assoc}})}{\sqrt{V_{n\Delta}(p_T^{\text{assoc}}, p_T^{\text{assoc}})}}, n = 2, 3 \quad (9)$$

432 Here, the reference particle p_T^{ref} range is chosen to be $0.3 < p_T < 3.0 \text{ GeV}/c$.

6.2 J/ψ v_n extraction

This analysis implement the similar technique to extract J/ψ v_n as described in AN-16-084 (HIN-16-007) and AN-17-223 (HIN-17-003), involving fit to the mass spectrum and v_n as function of invariant mass.

After v_n of all J/ψ candidates are measured as function of invariant mass, to extract v_n of J/ψ signal (v_n^{sig}), fits on mass spectrum and v_n as function of invariant mass is performed. The mass spectrum fit function is composed of two components: an exponential function to model the combinatorial background ($Bkg(m_{inv})$), a double Crystal Ball function for J/ψ signal ($Signal(m_{inv})$), as was done in AN-16-067 (HIN-16-004). The average v_n of all J/ψ candidates as a function of invariant mass, $v_n^{Sig+Bkg}(m_{inv})$, is fitted with Equation 10 and 11.

$$v_n^{Sig+Bkg}(m_{inv}) = \alpha(m_{inv})v_n^{sig} + (1 - \alpha(m_{inv}))v_n^{Bkg}(m_{inv}) \quad (10)$$

$$\alpha(m_{inv}) = Signal(m_{inv}) / (Signal(m_{inv}) + Bkg(m_{inv})) \quad (11)$$

$v_n^{Bkg}(m_{inv})$ is the v_n of background J/ψ candidates and is modelled as a exponential function of invariant mass. $\alpha(m_{inv})$ is the J/ψ signal fraction as a function of invariant mass, which is from mass spectrum fit function. Systematic uncertainties from v_n fit is evaluated in Sec. 8.3.

Figure. 26 shows fit on mass spectrum and v_2 as function of invariant mass in different p_T bins for $185 \leq N_{trk}^{offline} < 250$.

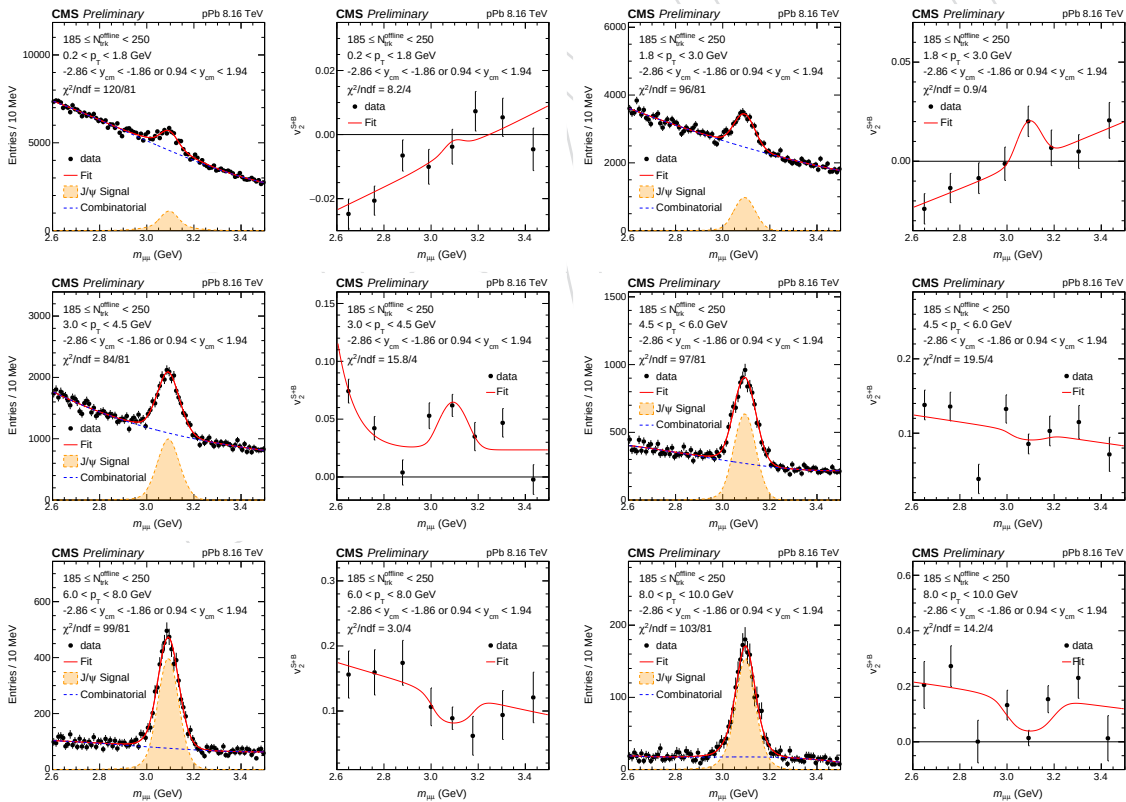


Figure 26: Fit on mass spectrum (left) and v_2 as function of invariant mass (right) in different p_T bins for $185 \leq N_{trk}^{offline} < 250$.

6.3 low multiplicity subtraction

The low multiplicity subtraction method developed in Refs. [17, 45] is employed in this analysis to estimate residual contribution from back-to-back jet correlations to v_2 results in pPb collisions. The v_2 results after subtracting correlations obtained from low-multiplicity pPb events are denoted as v_2^{sub} . The following paragraph provides brief description of the low multiplicity subtraction procedure, while more details can be found in AN-16-037 (HIN-16-010).

By assuming the jet-induced correlations are invariant with event multiplicity, the Fourier coefficients, $V_{n\Delta}$, extracted from Eq. 8 for $N_{\text{trk}}^{\text{offline}} < 35$ are subtracted from the data in the higher multiplicity region:

$$V_{n\Delta}^{\text{sub}} = V_{n\Delta} - V_{n\Delta}(N_{\text{trk}}^{\text{offline}} < 35) \times \frac{N_{\text{assoc}}(N_{\text{trk}}^{\text{offline}} < 35)}{N_{\text{assoc}}} \times \frac{Y_{\text{jet}}}{Y_{\text{jet}}(N_{\text{trk}}^{\text{offline}} < 35)}, \quad (12)$$

where Y_{jet} represents the near-side jet yield which is defined as integral of near side jet peak in the short range ($|\Delta\eta| < 1$) 1-D $\Delta\phi$ correlation functions. The ratio, $Y_{\text{jet}}/Y_{\text{jet}}(N_{\text{trk}}^{\text{offline}} < 35)$, is introduced to account for the enhanced jet correlations due to the selection of higher multiplicity events.

In this analysis, the subtraction is performed with $V_{2\Delta}^{\text{sig}}$ which is obtained in the same way as for the v_2 results, as described in Sec. 6.2.

6.3.1 Jet yield estimation for J/ Ψ -hadron correlation

The J/ ψ candidates used in this analysis have $1.4 < |\eta| < 2.4$, which is at the edge or even beyond the $|\eta| < 2.4$ acceptance of charged particles. Therefore, charged particles which are in the same jets as J/ ψ candidates can be outside the $|\eta| < 2.4$ acceptance, which leads to an in-complete near-side jet peak in the correlation functions, i.e. truncated.

The magnitude of this effect depends on the η value of J/ ψ candidates. Figure. 27 shows the average η of J/ ψ candidates in each p_T bin for events with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ and $N_{\text{trk}}^{\text{offline}} < 35$, as well as their ratios. No strong multiplicity dependence has been observed for the entire p_T region measured, which indicates jets are more or less truncated by the same amount at low and high multiplicity. Thus, when taking the ratio of jet yield as in Eq. 13, the effect from truncation are largely cancelled.

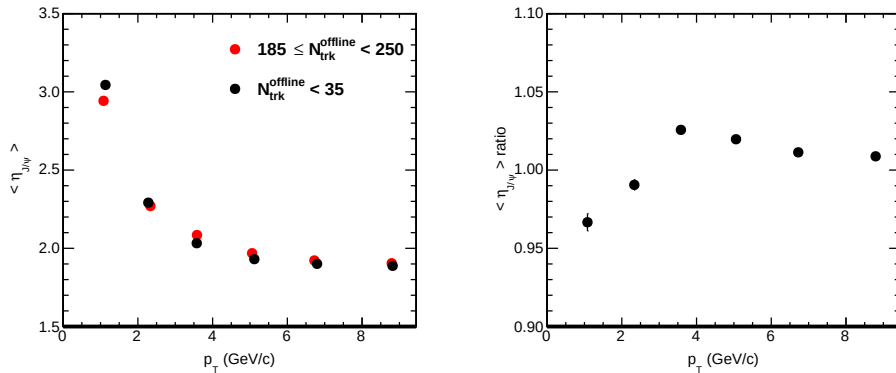


Figure 27: The average η of J/ ψ candidates in each p_T bin for events with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ and $N_{\text{trk}}^{\text{offline}} < 35$ (left), as well as their ratios (right).

To further ensure there is no large bias from truncated jet peak, jet yields are only extracted from correlation functions with J/ψ candidates have $p_T > 4.5$ GeV/c, where no obvious jet peak truncation has been observed in the near-side 1D projection of the correlation functions, as shown in Fig. 28 and Fig. 29 for low and high multiplicity.

To extract the jet yield for signal J/ψ , a side-band method is implemented. As shown in Fig. 28 and Fig. 29, jet yields are evaluated in the background region ($2.55 < m_{\mu\mu} < 2.94$ combined with $3.24 < m_{\mu\mu} < 3.63$) and the peak region ($2.94 < m_{\mu\mu} < 3.24$) of the mass spectrum. Signal J/ψ meson jet yield is then calculated by

$$Y_{\text{jet}}^{\text{sig}} = (Y_{\text{jet}}^{\text{peak}} - (1 - f)Y_{\text{jet}}^{\text{bkg}}) / f, \quad (13)$$

where $Y_{\text{jet}}^{\text{peak}}$ and $Y_{\text{jet}}^{\text{bkg}}$ are jet yield from peak and background region, and f is the J/ψ signal fraction extracted from the mass spectrum in the peak region. The jet yield ratio $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)}$ is then calculated as shown in Fig. 30, together with the same quantity for D^0 mesons (from HIN-17-003). Due to statistical limitations for J/ψ results and the observation that the same ratio for D^0 mesons has no strong p_T dependence, $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)}$ is fitted by a straight line for $p_T > 4.5$ GeV for J/ψ . The constant from the fitting is found to be 1.61 ± 0.18 as shown in Fig. 30. The subtracted results with $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)} = 1.61$ are reported as main results. To quoted systematic uncertainties for the subtraction, results are extracted with $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)}$ varied by 2 sigma (i.e. 1.61 ± 0.36) in $p_T > 4.5$ GeV region and $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)}$ varied by 3 sigma (i.e. 1.61 ± 0.54) in $p_T < 4.5$ GeV region. Figure. 31 shows v_2^{sub} results with different jet yield ratio, as well as the differences between them which are quote as systematic uncertainties. Table. 9 summarize the amount quoted.

Table 9: Summary of systematic uncertainties on v_2^{sub} from jet yield ratio for J/ψ in pPb collisions for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$.

p_T range (GeV/c)	Systematic uncertainty
0.2–1.8	0.0023
1.8–3.0	0.0070
3.0–4.5	0.0071
4.5–6.0	0.0066
6.0–8.0	0.011
8.0–10.0	0.014

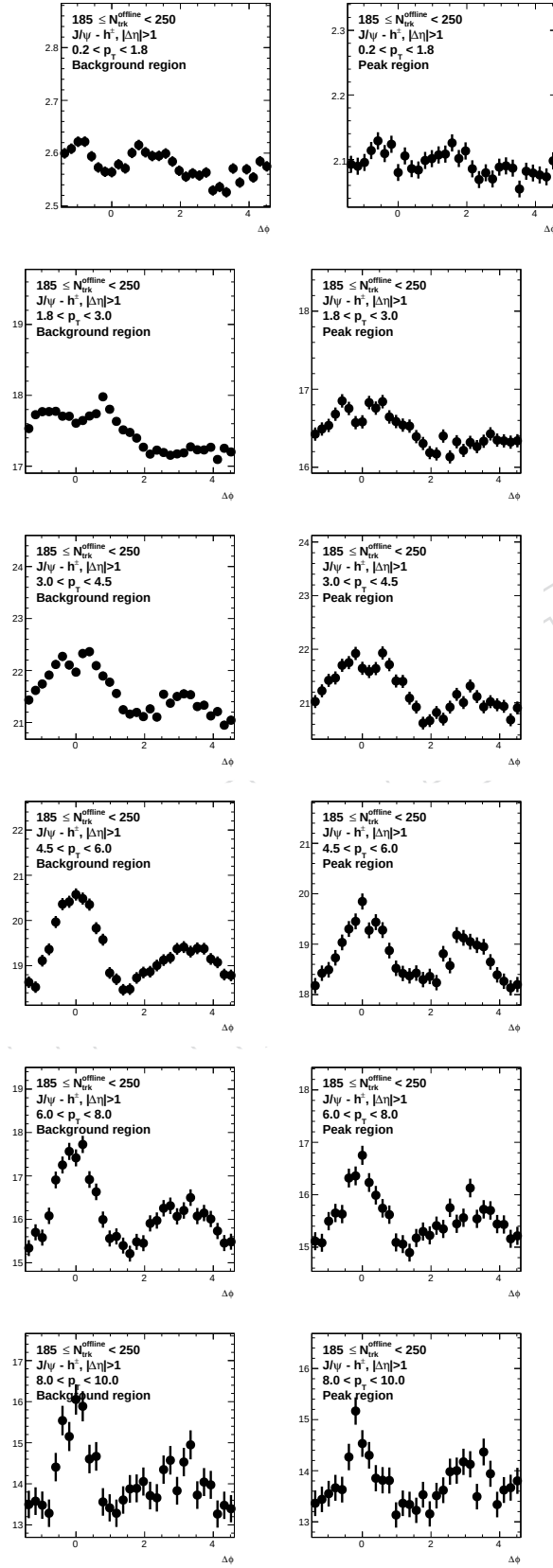


Figure 28: The near-side 1D projection of the J/ψ -hadron correlation functions for peak ($2.94 < m_{\mu\mu} < 3.24$) and background ($2.55 < m_{\mu\mu} < 2.94$) combined with $3.24 < m_{\mu\mu} < 3.63$ mass regions for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$.

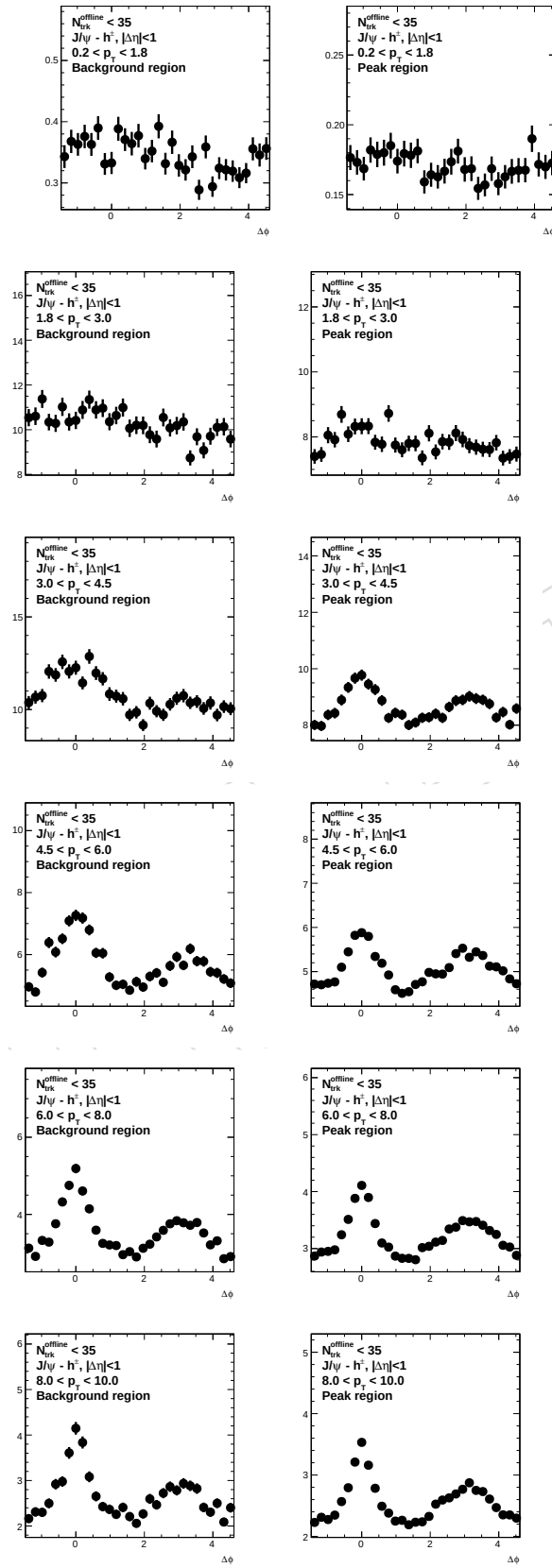


Figure 29: The near-side 1D projection of the J/ψ -hadron correlation functions for peak ($2.94 < m_{\mu\mu} < 3.24$) and background ($2.55 < m_{\mu\mu} < 2.94$) combined with $3.24 < m_{\mu\mu} < 3.63$ mass regions for $N_{\text{trk}}^{\text{offline}} < 35$.

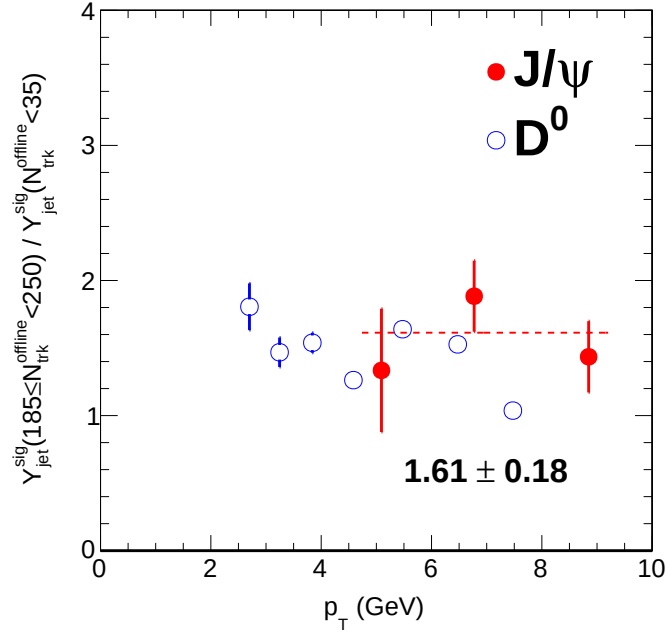


Figure 30: $\frac{Y_{\text{jet}}^{\text{sig}}}{Y_{\text{jet}}^{\text{sig}}(N_{\text{trk}}^{\text{offline}} < 35)}$ for J/ψ (left) and D^0 (right) mesons.

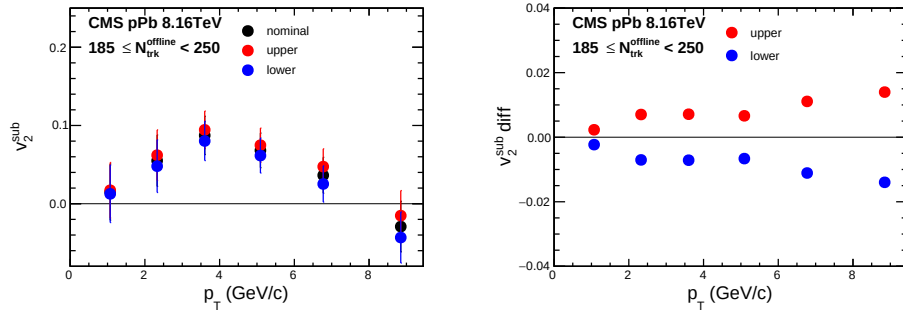


Figure 31: The v_2^{sub} results with different jet yield ratio, as well as the differences between results with jet yield ratio of 1 or 2 and results with jet yield ratio of 1.5.

7 Results

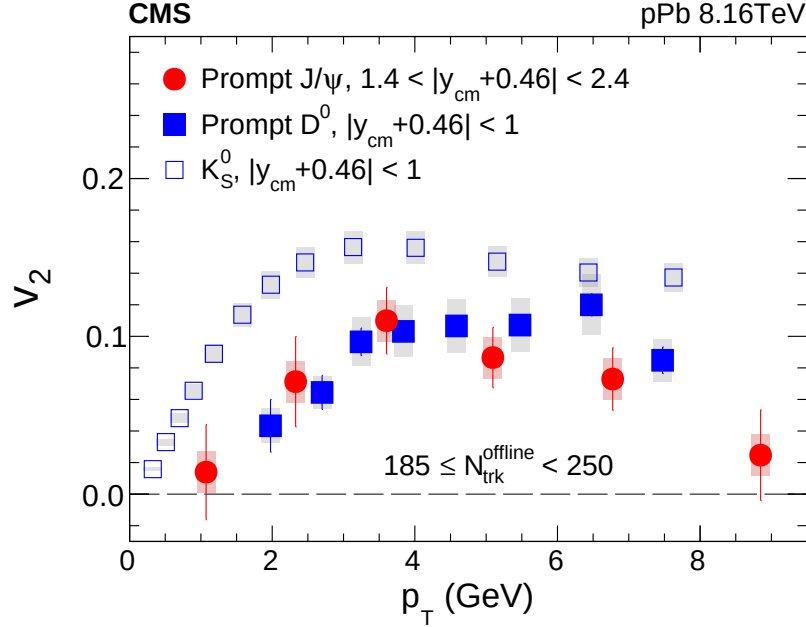


Figure 32: The v_2 results of the prompt J/ψ meson as a function of p_T in the multiplicity range $185 \leq N_{\text{trk}}^{\text{offline}} < 250$ of pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV. Data for K_S^0 and D^0 mesons from previous CMS measurements [46] are also shown for comparison. The error bars correspond to statistical uncertainties, while the shaded areas denote the systematic uncertainties.

Figure 32 shows the v_2 results of prompt J/ψ meson at forward rapidities ($-2.86 < y_{\text{cm}} < -1.86$ and $0.94 < y_{\text{cm}} < 1.94$) for high-multiplicity ($185 \leq N_{\text{trk}}^{\text{offline}} < 250$) pPb collisions. Significant positive values of prompt J/ψ v_2 are observed over a wide p_T range from about 2 to 8 GeV. The prompt J/ψ v_2^{sub} results show a trend of rising and declining with p_T . This observed trend appears to be in common with other meson species, such as K_S^0 (light, strange flavor) and D^0 (open heavy flavor) mesons, reported in the previous CMS publication [46] (where several strange baryon species are also studied but not shown in this paper). At low and intermediate p_T range of < 5 GeV, the v_2 values for J/ψ and D^0 mesons are consistent with each other within uncertainties, while an indication of a smaller v_2 signal for J/ψ mesons than that of D^0 mesons is seen at higher p_T for $p_T > 5$ GeV. Over the full p_T range, the v_2 signal for both J/ψ and D^0 mesons is smaller than that of K_S^0 mesons. This observation is consistent with earlier conclusion that charm quarks appear to develop a weaker collective dynamics than that of light quarks in small collision systems [46], unlike in AA collisions.

To better study the elliptic flow signal purely coming from long-range collective correlations, the v_2 results with residual jet correlations estimated and corrected for (v_2^{sub}) are shown in Fig. 33 (top) for prompt J/ψ mesons as well as for K_S^0 and D^0 mesons [46] as a function of p_T in pPb collisions with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. The effect of correction is most noticeable at very high p_T (although still not significant), while the overall p_T dependence of v_2 data remain unchanged. The correction for the v_2 of K_S^0 is larger (possibly because K_S^0 mesons are more coupled with the bulk multiplicity) and its v_2^{sub} value after the correction tend to converge with that of prompt J/ψ and D^0 at high p_T .

Motivated by the possible formation of a hydrodynamically expanding QGP medium in small systems, the elliptic flow signal for K_S^0 , J/ψ and D^0 mesons is compared as a function of trans-

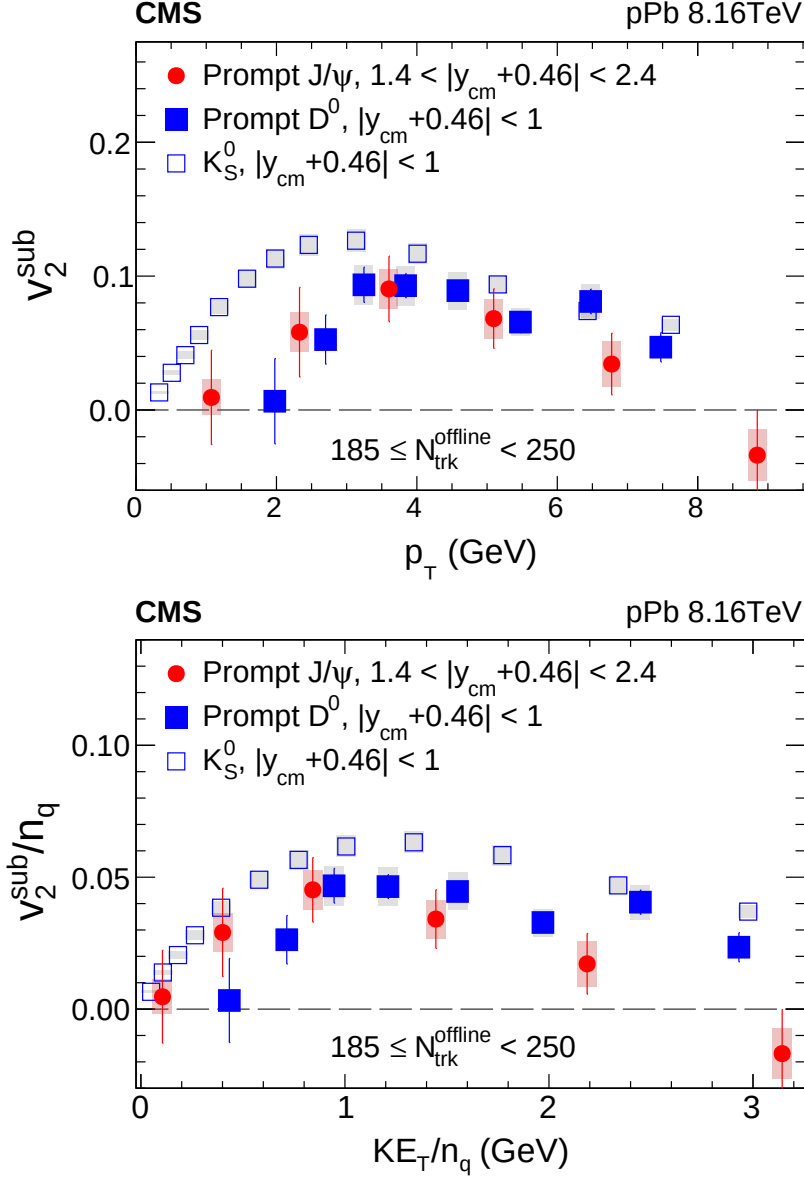


Figure 33: Top: the v_2^{sub} results for prompt J/ ψ meson, as well as for K_S^0 and prompt D^0 mesons, as a function of p_T for pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV with $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. Bottom: the n_q -scaled v_2^{sub} results. The K_S^0 and D^0 v_2 data are taken from Ref. [46]. The error bars correspond to statistical uncertainties, while the shaded areas denote the systematic uncertainties.

verse kinetic energy (KE_T) in Fig. 33 (bottom), to account for the mass difference among the three meson species [47, 48]. Here, the values of v_2^{sub} and KE_T are both divided by the number of constituent quarks, n_q , as shown in Fig. 33 (bottom), to represent collective flow signal at the partonic level, in the context of quark coalescence model [49–51]. A scaling of n_q -scaled v_2 values has been observed between light baryon and meson species in pA [23, 46] and AA [47, 48, 52] systems (known as the NCQ scaling), indicating that collectivity is first developed among partons, which later recombine into final-state hadrons. However, testing the NCQ scaling is not the main goal in this Letter so only three meson species are compared with a common n_q value of 2. Over the full KE_T/n_q range, the elliptic flow signal per quark for prompt D^0 and J/ ψ mesons is consistently below that of K_S^0 meson. For $KE_T/n_q > 1$ GeV, an indication of a smaller v_2 value of J/ ψ than that of D^0 is seen, which would again be consistent with a weaker collective

523 behavior of heavy flavor quarks than light quarks or gluons, possibly due to the much smaller
524 size of the collision system. However, no definitive conclusion can be drawn yet with current
525 experimental uncertainties.

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8 Systematic uncertainties and cross checks

Systematic uncertainties are studied for J/ψ v_2 results. The systematic sources include remaining non-prompt J/ψ , background v_2 PDF variation, tag-and-probe for muon trigger efficiency and J/ψ reconstruction efficiency correction. For each source of systematic effects investigated, differences of v_2 are calculated for various alternative conditions with respect to the default condition used in the analysis. When calculating the statistical uncertainties on the differences, only un-correlated uncertainties are taken into account. Since there is no p_T dependence observed, the differences observed are fitted by a straight line. The values from the fits, plus 1 sigma of their uncertainties, are quoted as systematic uncertainties. Systematic uncertainties from HLT trigger bias and effects from pileups are assumed to be the same as those for charged particles in HIN-16-022 (AN-16-370).

Table. 10 summarized different sources of systematic uncertainties on v_2 and v_2^{sub} results for J/ψ in pPb collisions for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$. The uncertainties from low multiplicity subtraction for v_2^{sub} are described in Sec. 6.3.1 and summarize in Table. 9. The total systematic uncertainties are evaluated by adding all the listed sources in quadrature as they are all independent, two significant digits are kept where the numbers are always rounded up. Since the systematics uncertainties for HLT trigger and pileup effects were quoted as percentage in HIN-16-022, the range of absolute values on the results are shown in Table. 10. Comparing to other systematic sources, HLT trigger and pileup effects are relatively very small, they are covered when the numbers are rounded up.

The detail discussion of each source is presented below.

Table 10: Summary of systematic uncertainties on v_2 for J/ψ in pPb collisions for $185 \leq N_{\text{trk}}^{\text{offline}} < 250$.

Source	
Non-prompt J/ψ	0.0042
Bkg v_2 PDF	0.0093
Efficiency correction	0.0077
MC closure	0.0003 - 0.0015 (1.5%)
HLT trigger bias	0.0002- 0.001 (1.0%)
Effect from pileups	0.0001 - 0.0005 (0.5%)
Total	0.013

8.1 Systematic uncertainties from background PDF variation in J/ψ signal extraction

The J/ψ mass spectrum fitting provides signal fraction which affect the determination of v_2 results. To test the robustness of the fitting, 2nd order and 3rd order polynomial background PDF were used instead of exponential to check the uncertainties. Fig. 34 shows variation in v_2 and difference from the default v_2 results. No systematic uncertainty has been quoted as the differences are consistent with 0.

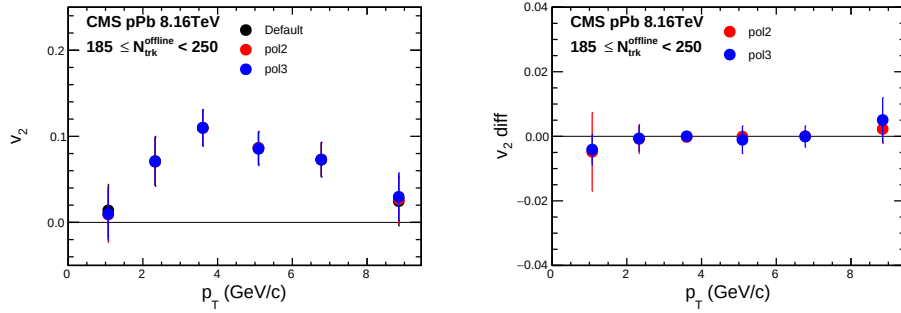


Figure 34: Variation in v_2 (left) and difference from the default v_2 results (right).

8.2 Systematic uncertainties from signal PDF in J/ψ signal extraction

The J/ψ mass spectrum fitting provides signal fraction which affect the determination of v_2 results. As described in Sec. 5.1, the signal Crystal Ball functions have two parameters (α and n) fixed from MC simulation, the robustness of the fitting has been tested by releasing the two parameters one-by-one. Fig. 35 shows variation in v_2 and difference from the default v_2 results. No systematic uncertainty has been quoted as the differences are consistent with 0.

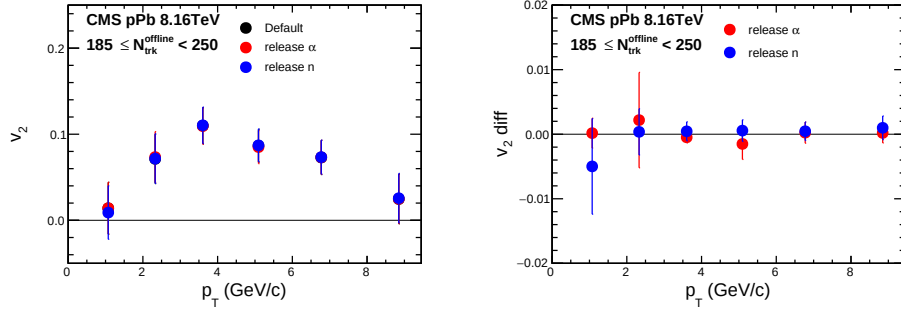


Figure 35: Variation in v_2 (left) and difference from the default v_2 results (right).

8.3 Systematic uncertainties from background v_2 PDF variation

The fit to $v_2(m_{inv})$ is sensitive to the background v_2 PDF. To test the goodness of the assumption that background v_2 is a exponential function of invariant mass, background $v_2(m_{inv})$ is assumed to be linear function, 2nd order polynomial (pol2) and 3rd order polynomial (pol3). Fig. 36 shows variation in v_2 and difference from the default v_2 results. Fig. 37 shows the v_2 fit using linear background function. Fig. 38 shows the v_2 fit using 2nd order polynomial background function. Fig. 39 shows the v_2 fit using 3rd order polynomial background function. Systematic uncertainty of 0.0093 is quoted.

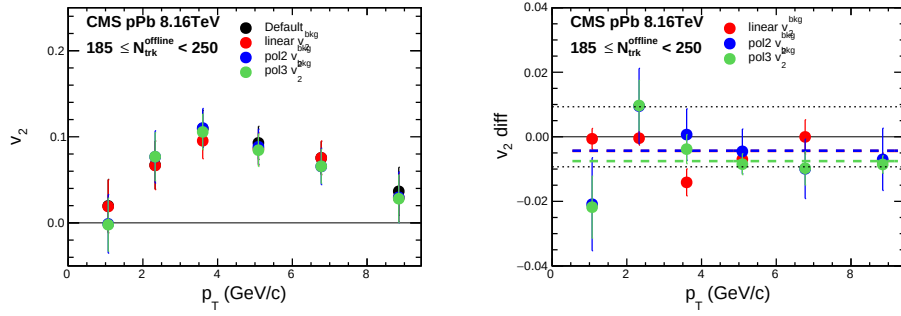


Figure 36: Variation in v_2 (left) and difference from the default v_2 results (right). The colored dash lines are the straight line fit to the differences with same color. The black dash line indicate the systematic uncertainty quoted.

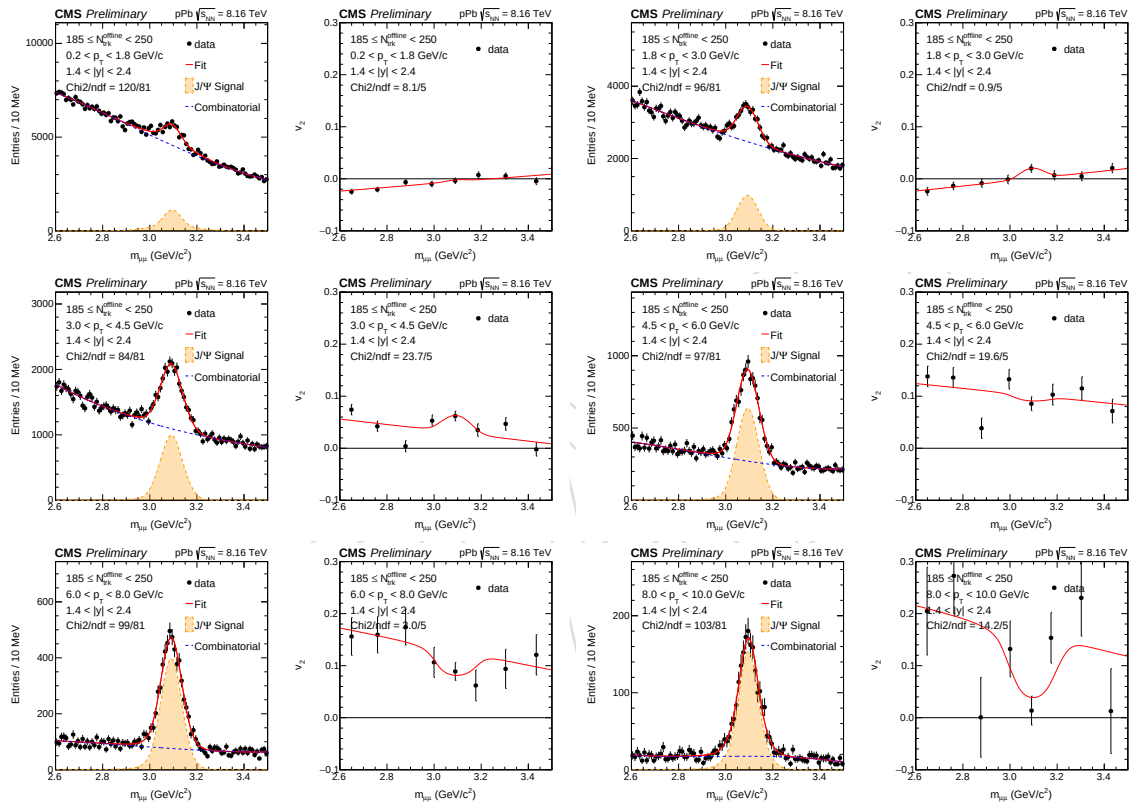


Figure 37: Fit on mass spectrum (left) and v_2 as function of invariant mass (right) in different p_T bins for $185 \leq N_{\text{offline}}^{\text{trk}} < 250$ using linear background function for v_2 .

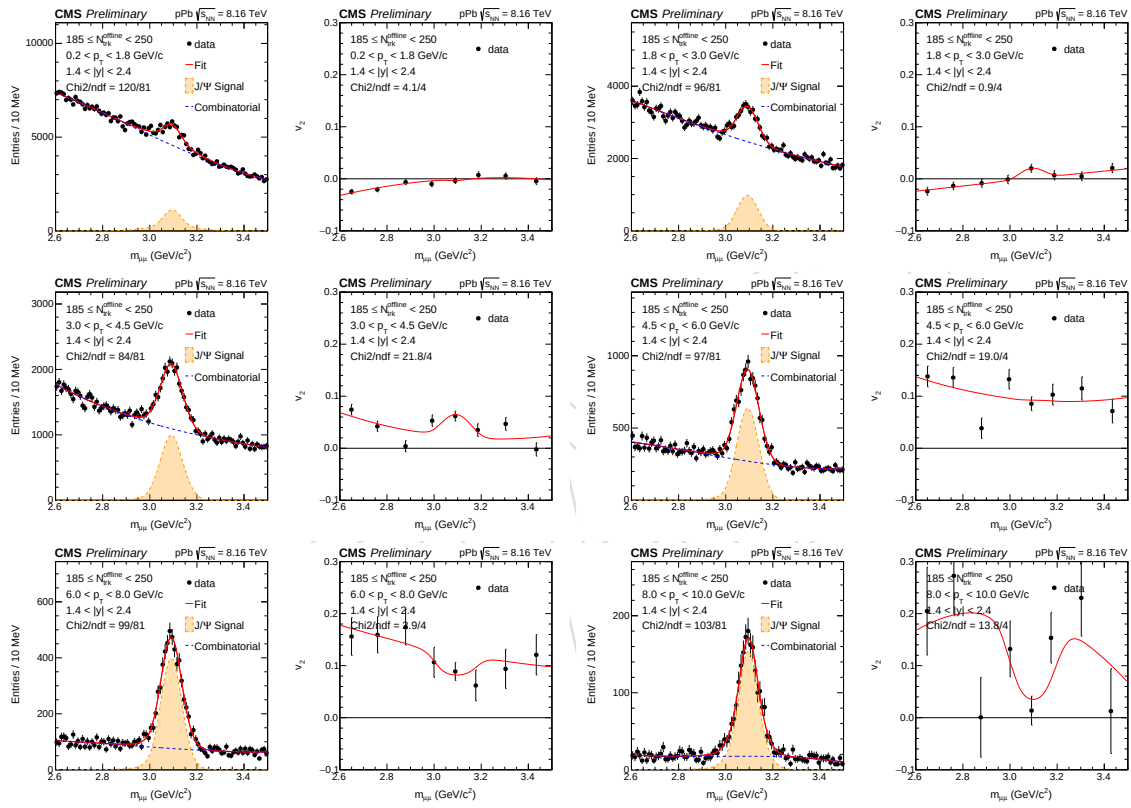


Figure 38: Fit on mass spectrum (left) and v_2 as function of invariant mass (right) in different p_T bins for $185 \leq N_{\text{offline trk}} < 250$ using 2nd order polynomial background function for v_2 .

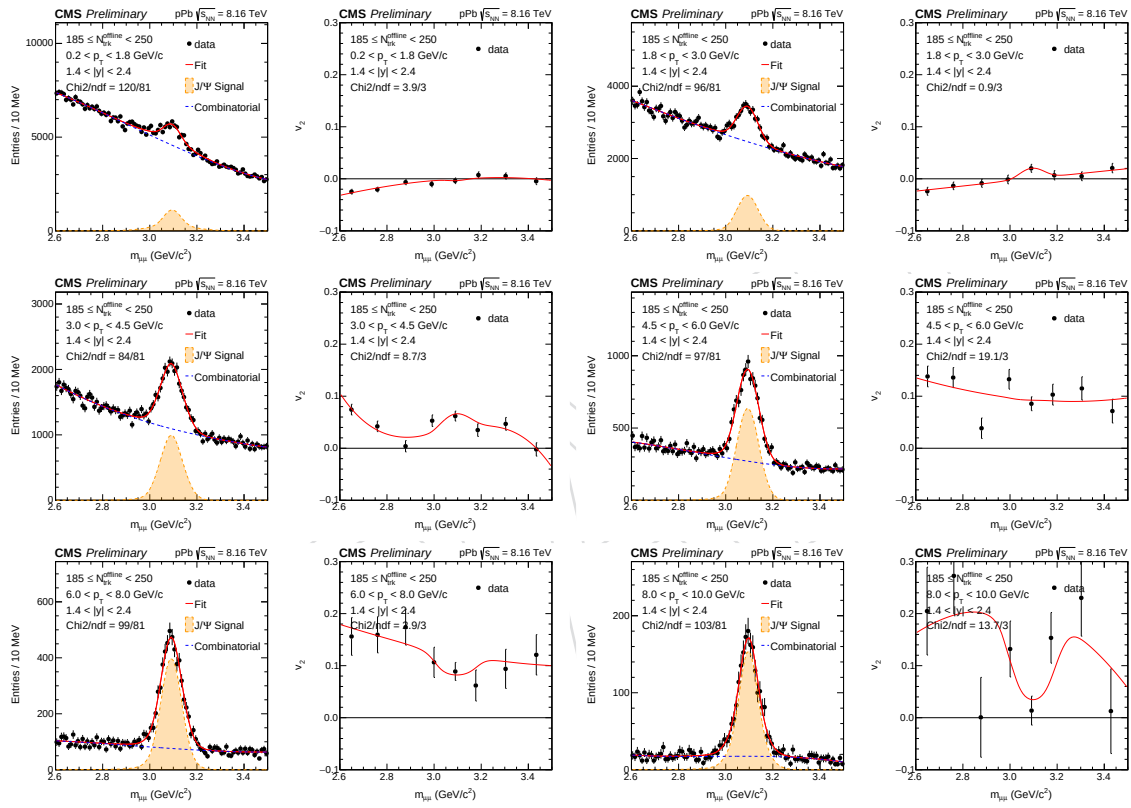


Figure 39: Fit on mass spectrum (left) and v_2 as function of invariant mass (right) in different p_T bins for $185 \leq N_{\text{offline trk}} < 250$ using 3rd order polynomial background function for v_2 .

8.4 Systematic uncertainties from J/Ψ efficiency correction

To evaluate the systematic uncertainty coming from efficiency correction, v_2 results are measured without any efficiency correction. Comparing to the default result as shown in Figure 44, a difference of 0.0077 is seen which is quoted as the systematic uncertainty.

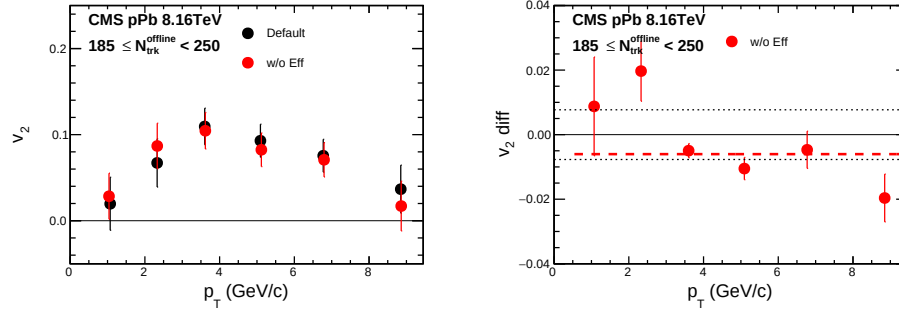


Figure 40: Variation in v_2 (left) and difference from the default v_2 results (right). The colored dash lines are the straight line fit to the differences with same color. The black dash line indicate the systematic uncertainty quoted.

8.5 Systematic uncertainties from non-prompt J/Ψ

The non-prompt J/ψ rejection, as described in Sec. 5.2, rejects 75% to 90% of all non-prompt J/ψ. To estimate the effect of the residual non-prompt J/ψ, v_2 results are measured with a looser non-prompt J/ψ rejection, which increases the number of non-prompt J/ψ passing the cut by a factor of 2. The looser cut is shown in Fig. 41. Fig. 42 shows variation in v_2 and difference from the default v_2 results, where 0.0042 is quoted as the systematic uncertainty.

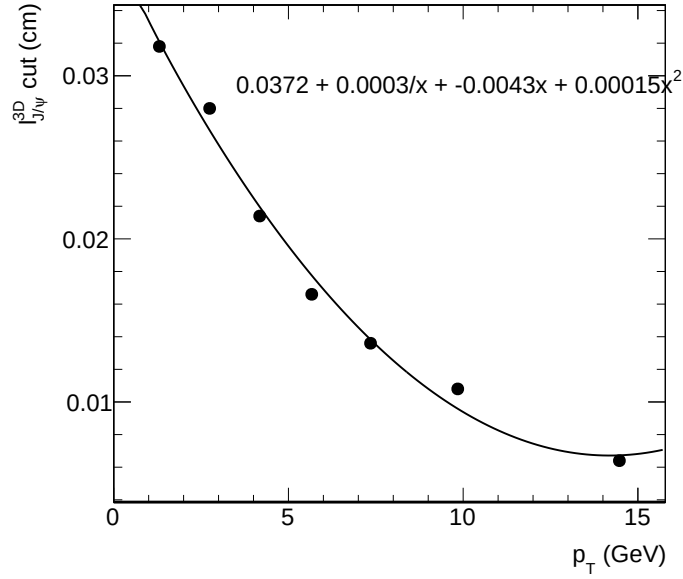


Figure 41: The dependence with p_T of the $\ell_{J/\psi}^{3D}$ cut values which increase the number of non-prompt J/ψ passing the cut by a factor of 2, for $1.4 < |y| < 2.4$. The cut values are fitted by a function $(a + b/p_T + c * p_T + d * p_T^2)$. Fit parameters are also shown.

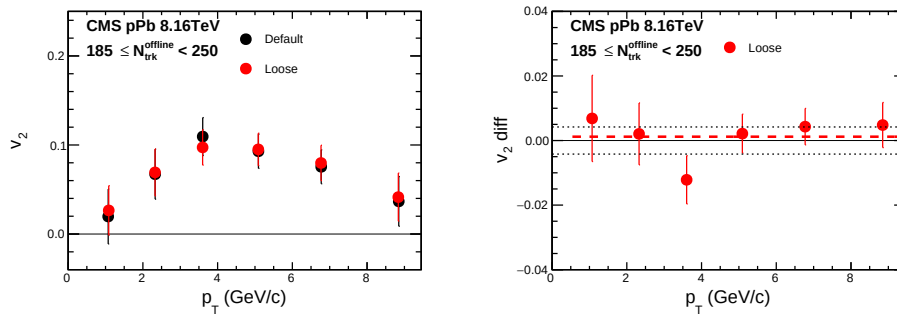


Figure 42: Variation in v_2 (left) and difference from the default v_2 results (right). The colored dash lines are the straight line fit to the differences with same color. The black dash line indicate the systematic uncertainty quoted.

8.6 Systematic uncertainties from Tag-and-Probe (TnP) correction

In this analysis, the efficiencies for J/ψ are computed from MC simulations. There are possible systematic uncertainties from discrepancies between the single muon trigger efficiencies computed in MC and those in data. In order to evaluate such uncertainties, the tag-and-probe (TnP) scale factors have been applied to J/ψ daughter muons when using DoubleMu triggered data samples (in this analysis only for $N_{\text{trk}}^{\text{offline}} < 35$).

To estimate the effect, the TnP scale factors are taken from AN-16-084 (from Fig.49) for pp 5 TeV L1DoubleMu0 trigger. The J/ψ $V_{2\Delta}$ results with TnP scale factors are compared to those without scale factor in Fig. 43. No discrepancy has been observed, thus no systematic uncertainty is assigned.

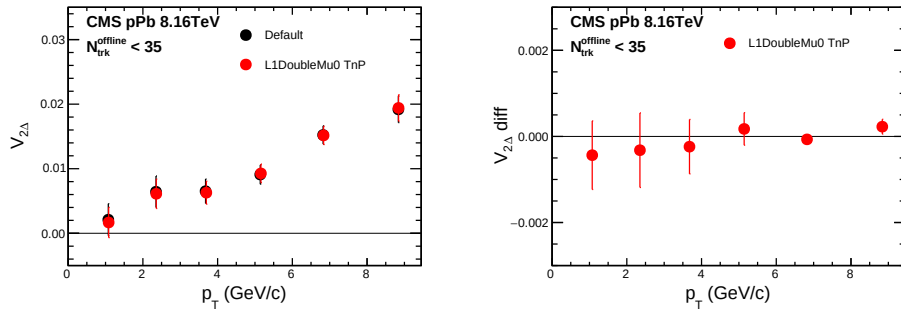


Figure 43: Comparison of J/ψ $V_{2\Delta}$ results with and without TnP scale factor for $N_{\text{trk}}^{\text{offline}} < 35$ from DoubleMu triggered samples.

8.7 Monte Carlo closure test

To ensure our analysis can reproduce generator level truth in Monte Carlo, a closure check is done using PYTHIA J/ ψ signal sample. v_2 values for J/ ψ are measured for generator level and reco level. The test are sensitive to all systematic uncertainties from the procedure of extracting the J/ ψ signal v_2 and also from the efficiency corrections. A systematic uncertainty of 1.5% is observed, which is negligible compare to other sources.

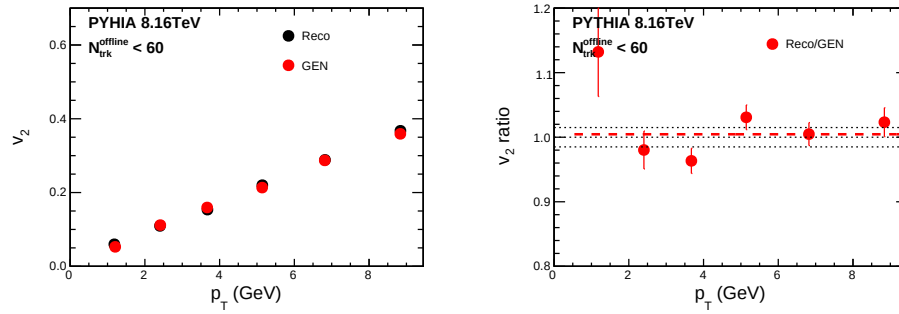


Figure 44: Comparison (left) and ratio (right) of J/ ψ v_2 results for PYTHIA generator level and reco level as a function of p_T . The colored dash lines are the straight line fit to the differences with same color. The black dash line indicate the systematic uncertainty quoted.

8.8 Cross check on v_2 statistical uncertainty from fit

As described in Sec. 6.2, the signal v_2 values for J/ψ are extracted from fits to v_2 distributions as functions of invariant mass. To check if the statistical uncertainties returned from the fits are correct, a test has been performed using smeared v_2 distribution as function of invariant mass. Based on the fit function for the nominal v_2 distributions, smeared v_2 values are generated. For each invariant mass bin, v_2 is evaluated from the nominal fit function and smeared according to the statistical uncertainty from data, assuming Gaussian distribution. The smeared v_2 distributions are then fitted to extract signal v_2 values for J/ψ . The smearing has been done for 100k times for each p_T bin. The extracted J/ψ signal v_2 distributions are shown in Fig. 45 and fitted by Gaussian function. Table 11 shows σ of the Gaussian fits and the statistical uncertainties from nominal v_2 results. No large discrimination has been observed between the two quantities, thus the statistical uncertainties on the nominal results are correct.

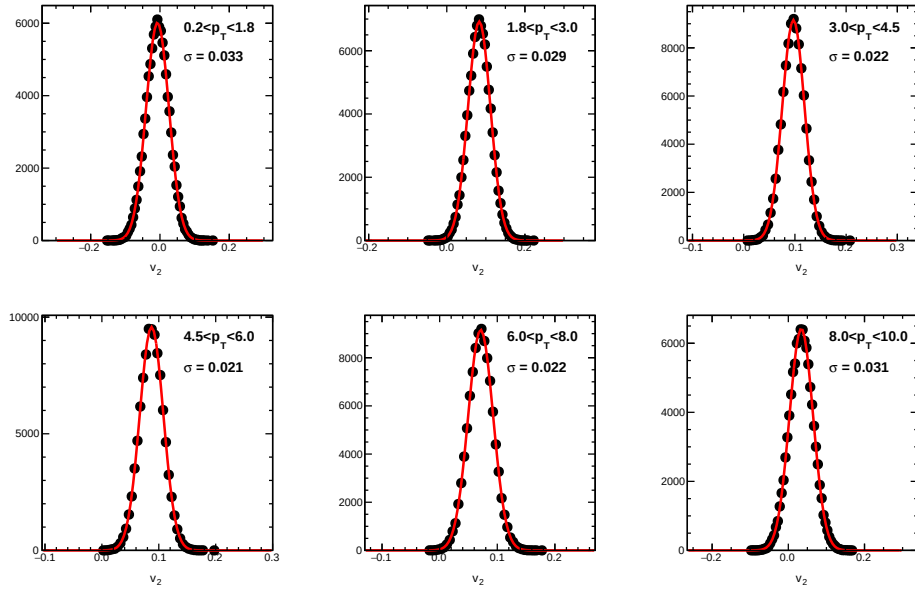


Figure 45: The extracted J/ψ signal v_2 distributions from the smearing test.

Table 11: Comparison between σ of smeared v_2 distribution and statistical uncertainties from nominal result.

p_T range	σ of Gaussian fit	Statistical uncertainty from nominal result
0.2–1.8	0.033	0.033
1.8–3.0	0.029	0.029
3.0–4.5	0.022	0.022
4.5–6.0	0.021	0.020
6.0–8.0	0.022	0.021
8.0–10.0	0.031	0.030

9 Summary

In summary, measurement of elliptic flow for prompt J/ψ meson in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV are presented. Significant positive v_2 values are observed for prompt J/ψ at forward rapidity ($1.4 < |y| < 2.4$) over a wide p_T range ($2 < p_T < 8$ GeV). This observation provides unambiguous evidence for the presence of collectivity for charm quarks in high-multiplicity pPb collisions, as for that first observed for light flavor hadrons. The observed ordering of v_2 among light flavor, open heavy flavor and J/ψ mesons at intermediate and high p_T region (>4 GeV) further support the earlier conclusion that heavy quarks exhibit weaker collective behavior than light quarks in small collision systems, unlike in AA collisions. New results presented, together with data for light and open heavy flavor hadrons, provide strong constraints on the dynamics of heavy quarks produced in small hadronic collision systems with high multiplicities.

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